A Biologically Grounded Virtual Zebrafish: From Retinal Sampling to Active Inference Policy Selection

Technical Report — Zebrafish SNN v1 (Steps 1–45)

Hae-Jeong Park
Department of Nuclear Medicine & Cognitive Science
Yonsei University College of Medicine, Seoul, Korea
parkhj@yonsei.ac.kr

March 27, 2026

Abstract

We present a spiking neural network (SNN) model of the larval zebrafish brain that implements the full perception–action loop: from binocular retinal sampling through tectal processing and predictive coding to locomotor output with neuromodulatory control. The model is developed in forty-five progressive steps, each adding a biologically motivated computational module. Steps 1–6 establish visual tracking with precision adaptation, dopamine-based reward prediction error, basal ganglia motor gating, optic tectum dynamics, and cross-modal salience. Steps 7–10 close the sensorimotor loop with autonomous locomotion, genomic (evolutionary) pre-training, and Hebbian fine-tuning. Steps 11–12 add object classification and classification-guided reactive behavior. Step 13 transitions from reactive to deliberative behavior via Expected Free Energy (EFE)-based policy selection with working memory, implementing three goal states—FORAGE, FLEE, and EXPLORE. Step 14 embeds the brain in a Gymnasium-compatible environment with energy homeostasis, depth-shaded vision, obstacle collision physics, and a habit network that consolidates stable EFE decisions into fast shortcut responses. Step 15 adds a reinforcement learning critic that adapts EFE weights online via temporal-difference error. Step 16 introduces a variational autoencoder (VAE) world model with associative memory and forward-simulation planning. Step 17 implements hippocampal place cells with path integration, replay, and spatial planning. Step 18 adds allostatic interoception—predictive tracking of hunger, fatigue, and stress that biases goal selection. Step 19 provides a full evaluation suite comparing baseline, full-brain, and place-cell configurations. Step 20 introduces an amygdala fear circuit with persistent threat arousal, flee energy cost asymmetry, and nonlinear hunger-driven foraging urgency. Step 21 adds active-inference food prospection with line-of-sight reachability scoring, olfactory heading cues, and retinal obstacle navigation. Step 22 introduces a curriculum reinforcement learning framework with progressive difficulty scaling. Step 23 adds a sleep–wake cycle for memory consolidation with precision freeze and fatigue recovery. Step 24 completes the active inference architecture by replacing all ground-truth environment reads with generative-model inference: a retinal feature extractor, VAE-based threat decoder, interoceptive energy model, and boundary-corrected path integration ensure the agent perceives only through its own sensory channels. Step 25 replaces single-compartment neurons with two-compartment predictive coding cells featuring apical dendrite feedback and goal-driven additive attention, with homeostatic gain control and PE-driven anti-Hebbian feedback learning, reducing prediction error 64% and improving survival from 100% to 60% catch rate. Step 26 adds a Bayesian survival trade-off that weighs starvation risk against predator threat in goal selection, an enhanced classifier with aggregate pixel-count features achieving 100% accuracy across all five entity classes, and a structured online feedback weight training protocol with frozen feedforward path that improves foraging 133% (6→14 food). All modules use biologically plausible local learning rules without backpropagation

through time. Steps 27–28 add ecological realism: multi-size food with density-based optimal foraging (marginal value theorem), binocular depth estimation from stereo overlap, adaptive predator with chase acceleration, motor primitive repertoire with burst-glide-idle bout dynamics, Mauthner cell C-start escape reflex triggered by looming detection (ℓ/v ratio), and three-phase prey capture kinematics. Step 29 evaluates decision rationality via structured scenarios, achieving 72/100 ("rational") on five cognitive dilemmas. Step 30 completes curriculum reinforcement learning with a Bayesian starvation trade-off resolving the dual predator-hunger threat. Step 31 replaces the monolithic VAE world model with three domain-specific generative models: a geographic model (40×30 grid of obstacle occupancy and food density), a Kalman-filter predator belief model with object permanence, and an interoceptive metabolic simulator enabling counterfactual active inference over 30-step energy trajectories. Steps 32–36 add peripheral sensorimotor systems: a 16-neuromast lateral line with reafference cancellation (Step 32), a Marr–Albus–Ito cerebellar forward model with LTD-based motor correction (Step 33), a 5-fish multi-agent environment with confusion-effect shoaling (Step 34), an olfactory chemotaxis and alarm-substance circuit (Step 35), and a habenula disappointment-driven strategy-switch module (Step 36). Steps 37–42 add vestibular otolith sensing with VOR (Step 37), a 32-neuron spiking spinal CPG implementing a half-centre oscillator with V2a excitatory and V0d inhibitory interneurons (Step 38), four-cone-type color vision with UV/B/G/R opponent channels (Step 39), a 2000-step circadian clock with crepuscular foraging peaks (Step 40), proprioceptive command–actual prediction error with collision detection (Step 41), and an insula module integrating heart rate, arousal, and emotional valence (Step 42). Step 43 adds three-factor reward-modulated STDP [48] to all deep feedforward synapses, reviving the quiescent PC_int layer (RMS $0.002 \rightarrow >0.01$) and implementing a biologically plausible reinforcement learning rule within the predictive coding hierarchy. Steps 44–45 refine behavioral precision: Step 44 improves ground-truth danger-zone detection and detour planning; Step 45 fixes the flee orbit bug and introduces targeted goal-arbitration interventions (PATROL predator bias, food-visible FORAGE reflex, STALK lateral avoidance, HUNT sprint continuity) that raise overall decision quality to 89/100 RATIONAL (A=100, B=85, C=80, D=100, E=79). In closed-loop simulations, the virtual zebrafish successfully forages for food, avoids predators with biologically realistic escape responses, navigates around rock formations using motor primitives, and explores its environment with goal-appropriate behavioral modulation driven by interoceptive and exteroceptive signals.

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1 Introduction

Larval zebrafish (*Danio rerio*, 5–7 days post-fertilization) offer a unique model system: a vertebrate brain small enough for whole-brain calcium imaging, yet complex enough to exhibit visually guided prey capture, predator avoidance, and social behavior [1, 2].

This report documents a computational model—**ZebrafishSNN**—that recapitulates key neural circuits of the larval zebrafish using a hierarchy of spiking neural network layers. The model is built incrementally across twenty-nine developmental steps, each introducing one functional module. The design philosophy is:

1. **Biological plausibility**: leaky integrator neurons, local Hebbian plasticity, no backpropagation through time.
2. **Modularity**: each brain region (retina, optic tectum, pretectum, basal ganglia, thalamus) is an independent module.
3. **Active inference**: behavior is driven by minimizing expected free energy over a small goal space.
4. **Developmental curriculum**: modules are added and tested progressively, mirroring biological maturation.

2 Model Architecture

2.1 Overview

The full processing pipeline is:

$$\underbrace{\text{Retina}}_{800\times 2} \xrightarrow{\text{topographic}} \underbrace{\text{OT}_{L/R}}_{600\times 2} \xrightarrow{\text{lateral inh.}} \underbrace{\text{OT}_F}_{800} \xrightarrow{\pi_{\text{OT}}} \underbrace{\text{PT}}_{400} \to \underbrace{\text{PC}_{per}}_{120} \xrightarrow{\pi_{\text{PC}}} \underbrace{\text{PC}_{int}}_{30} \to \begin{cases} \text{Motor} & (200) \\ \text{Eye} & (100) \\ \text{DA} & (50) \end{cases}$$

A separate classification head branches from the bilateral retinal type channels, augmented with aggregate pixel-count features (Section 23.1):

$$[\mathbf{L}_{400:},\ \mathbf{R}_{400:},\ \tilde{n}_{\text{obs}},\ \tilde{n}_{\text{ene}},\ \tilde{n}_{\text{food}},\ \tilde{n}_{\text{bnd}}] \xrightarrow{804\to 128\to 5} \text{class logits.}$$

2.2 Neuron Model: Two-Compartment Leaky Integrator

Each layer uses a first-order discrete leaky integrator:

$$\mathbf{v}(t) = \tau\,\mathbf{v}(t-1) + (1-\tau)\,\big(\mathbf{x}(t)\,\mathbf{W}\big), \quad \tau = 0.8, \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{n_{in}\times n_{out}}$ is initialized as $0.01\,\mathcal{N}(0,1)$. The effective time constant is $\tau/(1-\tau) = 4$ steps. In Step 25 (Section 22), trainable layers are upgraded to two-compartment predictive coding neurons with apical feedback and attention modulation; the frozen topographic layers $\text{OT}_\text{L}/\text{OT}_\text{R}$ retain this single-compartment form.

2.3 Two-Compartment Neuron: Predictive Coding Upgrade (Step 25)

Step 25 replaces single-compartment leaky integrators in all trainable layers with *two-compartment predictive coding neurons*. Each neuron has a somatic compartment $\mathbf{v}_s$ (bottom-up drive) and an apical dendritic compartment $\mathbf{v}_a$ (top-down feedback):

$$\boldsymbol{\varepsilon} = \mathbf{v}_a - \mathbf{v}_s, \quad F = \sum_l \tfrac{1}{2}\,\bar{\pi}_l\,\|\boldsymbol{\varepsilon}^{(l)}\|^2. \tag{2}$$

The apical–somatic mismatch $\boldsymbol{\varepsilon}$ is the prediction error, and the total free energy F is minimized by PE-driven feedback weight learning (anti-Hebbian, reward-independent). Homeostatic gain

control (divisive normalization at RMS target $v^* = 5$) prevents cascading activity collapse in deep layers.

2.4 Spiking Spinal CPG (Step 38)

The motor output is implemented by a 32-neuron spiking half-centre oscillator. Each side has 8 V2a excitatory interneurons, 4 V0d commissural inhibitory neurons, and 4 motor neurons—all leaky integrate-and-fire (LIF) units:

$$v_i(t{+}1) = \tau_m\, v_i(t) + (1{-}\tau_m)(I_{\text{ext}} + I_{\text{syn}} + \xi), \tag{3}$$

with cross-inhibition $I_{\text{syn}} = w_{\text{exc}}\, \bar{s}_{\text{exc}} - w_{\text{inh}}\, \bar{s}_{\text{contra}}$ producing alternating L/R bursts. Speed $= (m_L + m_R)/2$; turn $= (m_R - m_L) \times 1.5$ (contralateral). The CPG glide phase is bypassed during FLEE (full escape speed).

2.5 Reticulospinal Motor Pathway (Step 41)

A direct crossed tectal→motor shortcut bypasses the signal-attenuated PC layers. Left OT (600-dim) projects to right motor neurons (100-dim) and vice versa—contralateral anatomy matching the tecto-bulbar pathway in zebrafish [46]. After offline supervised training on 15 903 flee episodes (MSE regression, Adam $\eta = 10^{-3}$, 10 epochs), the shortcut contributes up to 30% of the turn signal during normal navigation via confidence-weighted blending.

3 Sensory Processing

3.1 Binocular Retinal Sampling (Step 1)

Each eye samples a 20×20 angular grid (400 pixels) over a horizontal field of view of $\pm 80°$ and elevation range of $\pm 20°$. The left eye is centered at $\theta_{\text{head}} - 45°$, the right at $\theta_{\text{head}} + 45°$.

Each pixel has two channels:

- **Intensity channel** with Gaussian foveation:

$$G(\alpha) = \exp\Big(-\frac{\alpha^2}{2\sigma^2}\Big), \quad \sigma = 12°, \tag{4}$$

  where α is the azimuthal offset from eye center.
- **Type channel** (unfoveated) carrying entity identity: food $= 1.0$, obstacle $= 0.75$, enemy $= 0.5$, prey $= 0.38$, colleague $= 0.25$, boundary $= 0.12$, none $= 0.0$.

Total retinal output per eye: $\mathbf{L}, \mathbf{R} \in [0, 1]^{1\times 800}$.

3.2 Tectal Topography

Optic tectum weights are initialized with retinotopic Gaussian receptive fields:

$$W_{ij} = \exp\Big(-\frac{(x_i^{in} - x_j^{out} - \delta)^2}{2\sigma_T^2}\Big), \quad \sigma_T = 0.10, \tag{5}$$

where $\delta = -0.15$ (left) or $+0.15$ (right) implements nasotemporal shift. Type channels receive $0.7\times$ the intensity-channel weight.

3.3 Lateral Inhibition

Winner-take-all competition between left and right OT channels sharpens the bilateral contrast signal:

$$\mathbf{o}_L^{out} = \mathbf{o}_L - \lambda\, \mathbf{o}_R, \quad \mathbf{o}_R^{out} = \mathbf{o}_R - \lambda\, \mathbf{o}_L, \quad \lambda = 0.25. \tag{6}$$

3.4 Retinal Turn Signal

The basic turn signal is computed from intensity-channel asymmetry:

$$\text{turn} = \frac{\sum R_{1:400} - \sum L_{1:400}}{\sum R_{1:400} + \sum L_{1:400} + 10^{-8}}. \tag{7}$$

4 Predictive Coding and Precision (Step 2)

Under the Bayesian brain hypothesis [11, 14], the brain maintains a generative model of its sensory environment and minimizes *variational free energy*—an upper bound on surprise $\mathcal{S} = -\ln p(\mathbf{o})$—by adjusting both its internal model (perception) and its actions (active inference). Predictive coding [13] implements this minimization through a hierarchy of prediction errors, where each cortical layer predicts the activity of the layer below and updates its representation based on the mismatch [12].

4.1 Free Energy and Surprise

Variational free energy F upper-bounds Bayesian surprise:

$$\mathcal{S}(\mathbf{o}) = -\ln p(\mathbf{o}) \;\leq\; F = D_{\mathrm{KL}}\big(q(\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta}|\mathbf{o})\big) - \ln p(\mathbf{o}) = \underbrace{\mathbb{E}_q\big[-\ln p(\mathbf{o}|\boldsymbol{\theta})\big]}_{\text{prediction error}} + \underbrace{D_{\mathrm{KL}}\big(q(\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta})\big)}_{\text{complexity}}. \tag{8}$$

Under a Gaussian generative model with precision π, the prediction error term reduces to precision-weighted squared error:

$$F \approx \tfrac{1}{2} \sum_l \pi_l \left\|\boldsymbol{\varepsilon}^{(l)}\right\|^2 + \text{const.} \tag{9}$$

At the tectal fusion level (before Step 25 upgrades this to two-compartment PE): $F_{\mathrm{OT}} = \frac{1}{2}\,\overline{\pi}_{\mathrm{OT}}\,\|\mathbf{o}_{\text{fused}} - \mathbf{o}_F\,\mathbf{W}_{OTF}^{\top}\|^2$, where $\overline{\pi}_{\mathrm{OT}} = \text{mean}(\sigma(\boldsymbol{\gamma}_{\mathrm{OT}}))$.

4.2 Precision as Empirical Bayes

Precision π controls the gain of prediction errors: high precision amplifies errors (confident processing), low precision attenuates them (uncertain, exploratory processing). Under empirical Bayes, precision is itself inferred from data by optimizing free energy with respect to π, yielding $\hat{\pi} \propto 1/\mathrm{Var}[\varepsilon]$—the inverse of prediction error variance [28].

Each precision unit maintains a learnable logit $\boldsymbol{\gamma}$ with output $\pi = \sigma(\boldsymbol{\gamma})$, updated as:

$$\Delta\gamma = \alpha_\pi\,(|\varepsilon| - \beta) - 0.05\,\nu\,\mathbb{1}[\nu > 0.03], \tag{10}$$

where $|\varepsilon|$ is the mean absolute error, $\beta = 0.1$ is the target error level, $\nu = ||\varepsilon|(t) - |\varepsilon|(t{-}1)|$ is the novelty signal, and $\alpha_\pi = 0.05$. Precision is clamped: $\gamma \in [-3, 3]$.

The novelty term implements *Bayesian surprise* [29]: abrupt changes in prediction error signal a regime change in the generative model, triggering a precision dip that temporarily broadens the posterior and enables rapid model updating.

Two precision units operate at different hierarchical levels:

- π_{OT} (800-dim): applied to fused OT output $\mathbf{o}_F$.
- π_{PC} (120-dim): applied to perisomatic PC layer.

5 Neuromodulation

5.1 Dopamine System (Step 3)

The dopamine system models VTA/SNc dynamics with TD-style reward learning.

Pseudo-reward.

$$r(t) = \exp\big(-|F_{\text{vis}}(t)|\big) + 0.5 \cdot n_{\text{eaten}}. \tag{11}$$

Value learning and RPE.

$$\text{RPE}(t) = r(t) - V(t), \quad V(t{+}1) = \big(V(t) + \alpha_V \,\text{RPE}(t)\big) \cdot \delta_V, \tag{12}$$

with $\alpha_V = 0.05$ and $\delta_V = 0.98$.

Dopamine level.

$$\text{DA}(t) = \sigma\big(\beta_{\text{DA}} \cdot \text{RPE}(t)\big) + \xi, \quad \xi \sim \mathcal{N}(0, 0.01), \tag{13}$$

clamped to $[0, 1]$. Baseline DA $= 0.5$ (neutral); > 0.5 signals reward, < 0.5 signals aversion.

Bilateral valence.

$$\begin{aligned} v_L(t) &= (1{-}\delta_v)\, v_L(t{-}1) + 0.1\,(e_L - 0.5\, e_R), \\ v_R(t) &= (1{-}\delta_v)\, v_R(t{-}1) + 0.1\,(e_R - 0.5\, e_L), \end{aligned} \tag{14}$$

where $e_{L/R} = \overline{o}_{L/R} - 0.5\, V$, $\delta_v = 0.02$. Mean-subtracted and clamped to $[-0.5, 0.5]$.

Dopamine–precision coupling.

$$\Delta\gamma_{\text{OT/PC}} \mathrel{+}= 0.008\,(\text{DA} - 0.5). \tag{15}$$

5.2 Cross-Modal Salience — Thalamus (Step 6)

The thalamic relay computes cross-modal salience (CMS):

$$\text{CMS}(t) = (1{-}\alpha_s)\,\text{CMS}(t{-}1) + \alpha_s \Big[\gamma_\Delta \,\frac{\Delta F_v + \Delta F_a}{2} + (1{-}\gamma_\Delta)\,|F_v - F_a|\Big], \tag{16}$$

with $\gamma_\Delta = 0.6$, $\alpha_s = 0.1$.

CMS modulates downstream systems:

$$\begin{aligned} \beta_{\text{DA}} &= 3.0\,(1 + 0.5\,\text{CMS}), \qquad & \sigma_{\text{BG}} &= \sigma_0\,(1 + 2.0\,\text{CMS}), \\ \pi_{\text{bias}} &= \pi\,(1 - 0.2\,\min(\text{CMS}, 1)). \end{aligned} \tag{17}$$

6 Motor Control

6.1 Basal Ganglia Gating (Step 4)

The basal ganglia implements action selection through switching dynamics:

Drive. $d = (v_R - v_L) + 0.3\,(\text{DA} - 0.5)$.

Momentum integration.

$$p(t) = 0.65\, p(t{-}1) + 0.35\, d(t). \tag{18}$$

State update with dopamine-gated leak.

$$s(t) = s(t{-}1) + 0.3\, p(t) - \ell_{\text{eff}}\, s(t{-}1) + \sigma_{\text{BG}}\, \xi, \tag{19}$$

where $\ell_{\text{eff}} = 0.10\,(1 + 0.5\,\max(0,\ 0.5 - \text{DA}))$.

Gate output. $g = \tanh(0.8\, s) \in [-1, 1]$.

Exploratory switching. With probability $p_{\text{sw}} = 0.12\,\sigma(5\,|\text{RPE}|)$, momentum reverses sign, producing RPE-triggered saccadic switching.

6.2 Optic Tectum Eye Control (Step 5)

Continuous eye-movement dynamics integrating salience competition, free energy, and BG gating:

$$\begin{aligned} \dot{v}_{\text{eye}} &= 0.35\big[\tanh(1.5(v_R - v_L)) - 0.15\,F + 0.12\,g\big] - 0.35\,v_{\text{eye}}, \\ e(t{+}1) &= e(t) + 0.2\,v_{\text{eye}} - 0.18\,(1{+}0.5\max(0, 0.5{-}\text{DA}))\,e(t). \end{aligned} \tag{20}$$

6.3 Closed-Loop Locomotion (Step 7)

The complete locomotor command:

$$\begin{aligned} \theta(t{+}1) &= \theta(t) + 0.15\,\tilde{\tau} + 0.03\,g + 0.02\,e, \\ v_{\text{swim}} &= v_0\,s_{\text{mod}}\,(0.8 + 0.4\,\text{DA}), \\ x(t{+}1) &= x(t) + v_{\text{swim}}\cos\theta, \\ y(t{+}1) &= y(t) + v_{\text{swim}}\sin\theta, \end{aligned} \tag{21}$$

where $\tilde{\tau}$ is the smoothed, modulated turn signal, $v_0 = 1.5$, and s_{mod} is a goal-dependent speed multiplier.

7 Learning

7.1 Genomic Pre-training (Step 8)

Supervised learning mimics evolutionary pre-wiring of the motor pathway. Training uses diverse goal contexts (70% FORAGE, 20% FLEE, 10% EXPLORE) to build goal-conditioned motor responses:

$$\mathcal{L} = \|\hat{\tau} - s\,\tanh(2\alpha)\|^2 + 0.3\,\|\hat{\phi} - s\,\alpha/\pi\|^2, \tag{22}$$

where $\hat{\tau}$ is the decoded motor turn, $\hat{\phi}$ the predicted eye angle, α the angle to the entity, and $s = +1$ (approach for food) or $s = -1$ (avoid for enemy). Adam optimizer with lr $= 0.005$, cosine annealing, gradient clipping at 1.0, for 120 epochs $\times$ 50 samples.

7.2 Hebbian Fine-tuning (Step 10)

RPE-gated Hebbian plasticity refines connections during foraging:

$$\Delta\mathbf{W} = \eta\,\text{RPE}\,\text{DA}\,(\mathbf{x}_{\text{pre}}^{\top}\mathbf{x}_{\text{post}}), \quad \mathbf{W} \leftarrow 0.9999\,\mathbf{W}, \tag{23}$$

with $\eta = 0.0005$. Applied to OT fusion weights and motor output head.

7.3 Object Classification (Step 11)

A two-layer classifier (804→128→5) is trained on bilateral retinal type channels augmented with aggregate pixel counts using class-weighted cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = -\sum_c w_c\,y_c\log p_c, \quad p = \text{softmax}\big(\mathbf{W}_2\,\text{ReLU}(\mathbf{W}_1\,\mathbf{x}_{\text{cls}})\big), \tag{24}$$

where $\mathbf{x}_{\text{cls}}$ is the 804-dimensional input defined in Eq. 96 and $w_{\text{env}} = 3$ (other classes $w = 1$). Five classes: nothing, food, enemy, colleague, environment. Goal context is matched to entity type during training (e.g., FORAGE for food scenes, FLEE for enemy scenes). Trained for 250 epochs $\times$ 150 samples with lr $= 0.01$ and 8 integration steps for environment scenes; achieves 100% per-class validation accuracy.

8 Reactive Behavior (Step 12)

Classification probabilities directly modulate behavior:

$$\begin{aligned} a_{\text{gain}} &= 1.0 + 0.3\,p_{\text{food}} - 1.5\,\max(0,\ p_{\text{enemy}} - 0.6) + 0.15\,p_{\text{colleague}}, \\ s_{\text{mod}} &= 1.0 + 0.2\,p_{\text{food}} + 0.6\,\max(0,\ p_{\text{enemy}} - 0.5), \\ e_{\text{mod}} &= \max\big(0.2,\ 1.0 + 0.3\,p_{\emptyset} - 0.3\,(p_f + p_e)\big), \end{aligned} \tag{25}$$

where a_{gain} is the approach gain (positive = approach, negative = flee), s_{mod} is the speed multiplier, and e_{mod} is the exploration multiplier. The turn signal is modulated as $\tilde{\tau}_{\text{mod}} = \tau_{\text{raw}} \cdot a_{\text{gain}}$.

9 Active Inference Policy Selection (Step 13)

Step 13 transitions from frame-by-frame reactive behavior to deliberative goal selection via Expected Free Energy (EFE) minimization.

9.1 Goal Space

Four discrete goals: $\mathcal{G} = \{\textsc{forage},\ \textsc{flee},\ \textsc{explore},\ \textsc{social}\}$.

9.2 Expected Free Energy

For each goal g_i, the EFE combines epistemic (uncertainty-reducing) and pragmatic (reward-seeking) terms:

$$\boxed{\begin{aligned} G_{\textsc{forage}} &= 0.2\,\mathcal{U} - 0.8\,p_{\text{food}} + 0.2\,(0.5 - \text{DA}) + 0.15 - 0.6\,s^* + 0.15\,p_{\text{enemy}}, \\ G_{\textsc{flee}} &= 0.1\,\text{CMS} - 0.8\,p_{\text{enemy}} + 0.20 + 0.4\,s^*, \\ G_{\textsc{explore}} &= 0.3\,\mathcal{U} - 0.5\,(p_{\emptyset} + p_{\text{env}}) + 0.1\,\text{CMS} + 0.20 + 0.2\,s^*, \\ G_{\textsc{social}} &= 0.2\,\mathcal{U} - 0.6\,p_{\text{colleague}} + 0.15\,p_{\text{enemy}} + 0.25 + 0.15\,s^*, \end{aligned}} \tag{26}$$

where $\mathcal{U} = 1 - \frac{1}{2}(\pi_{\text{OT}} + \pi_{\text{PC}})$ is the model uncertainty. Lower EFE indicates a more preferred goal.

where s^* is the starvation risk from Eq. 97 (Step 26). Key design features:

- **Bayesian survival trade-off**: starvation risk s^* lowers forage EFE (urgency) while raising flee/explore/social EFE (cost). This creates a principled competition between predator avoidance and energy acquisition.
- Dopamine modulates $G_{\textsc{forage}}$: low DA increases foraging drive.
- CMS (novelty) promotes both fleeing and exploring.
- $G_{\textsc{social}}$ attracts toward conspecifics but is suppressed by predator presence and starvation.

9.3 Policy Posterior

The posterior over goals uses a softmax with inverse temperature $\beta = 2.0$:

$$\pi(g_i) = \frac{\exp(-\beta\,(G_i - G_{\min}))}{\sum_j \exp(-\beta\,(G_j - G_{\min}))}. \tag{27}$$

9.4 Persistence and Emergency Override

A timer-based persistence mechanism prevents goal flickering:

- Minimum persistence: 8 steps per goal episode.

- **Survival override**: if $p_{\text{enemy}} > 0.35$ and starvation urgency $s < 0.6$, force FLEE. If both threats are active ($p_{\text{enemy}} > 0.35$ and $s \geq 0.6$), defer to the EFE softmax posterior for Bayesian model comparison.
- **Starvation override**: if $s > 0.7$ with no immediate predator, force FORAGE.
- **Context-sensitive early exit**: FLEE exits early after 3 steps if $p_{\text{enemy}} < 0.20$; FORAGE exits after 4 steps if $p_{\text{food}} < 0.15$.
- Re-selection uses $\arg\max \pi(g)$.

9.5 Goal Arbitration Refinements (Step 45)

Four targeted interventions substantially improve decision quality:

PATROL predator bias. When the predator is in PATROL state and beyond 150 px, an EFE adjustment suppresses unnecessary fleeing and promotes foraging:

$$G_{\text{FORAGE}} \mathrel{-}= 10 \quad \text{(lower EFE = more preferred)},$$
$$G_{\text{FLEE}} \mathrel{+}= 10 \quad \text{(higher EFE = less preferred)}.$$

Without this correction, weak amygdala activation from a distant patrol predator caused 61–77% FLEE time in foraging scenarios.

Food-visible FORAGE reflex. When the retina detects food pixels ($n_{\text{food}} > 3$), the goal is immediately overridden to FORAGE (except during active FLEE):

$$g_{\text{eff}} = \begin{cases} \text{FORAGE} & n_{\text{food}} > 3 \ \wedge \ g \neq \text{FLEE}, \\ g & \text{otherwise}. \end{cases}$$

A 25-step post-eat lock-in maintains FORAGE after food consumption.

Starvation + STALK lateral avoidance. When critically starving ($E < 25\%$) and a STALK predator is within 200 px ahead, a lateral avoidance bias curves the fish perpendicular to the predator while still heading toward food (Eq. **??**).

HUNT sprint continuity. The flee sprint (1.5× speed) is extended to fire when a HUNT predator is within 300 px ground-truth distance, regardless of retinal occlusion (Eq. **??**).

9.6 Confidence (Meta-Precision)

Entropy-based confidence provides a meta-level precision signal:

$$\text{conf} = 1 - \frac{H(\boldsymbol{\pi})}{\ln 4}, \quad H(\boldsymbol{\pi}) = -\sum_i \pi(g_i) \ln \pi(g_i). \tag{28}$$

9.7 Goal-to-Behavior Mapping

Table 1: Behavioral modulation by active goal.

Goal	a_{gain}	s_{mod}	e_{mod}
FORAGE	$1.0 + 0.5\, p_f\, c$	$1.0 + 0.3\, p_f$	0.5
FLEE	$-(0.8 + 0.5\, p_e\, c)$	$1.3 + 0.4\, p_e$	0.3
EXPLORE	0.8	0.9	$1.5 + 0.5(1-c)$
SOCIAL	$0.6 + 0.3\, p_{\text{col}}\, c$	0.9	0.8

$p_f = p_{\text{food}}$, $p_e = p_{\text{enemy}}$, $c = \text{conf}$.

9.8 Working Memory

A 16-dimensional latent memory trace integrates classification, goal state, and neuromodulatory context:

$$\mathbf{z}_t = [p_1, \ldots, p_5,\ g_1, g_2, g_3,\ \mathrm{DA},\ \mathrm{CMS},\ F,\ \pi_{\mathrm{OT}},\ \pi_{\mathrm{PC}},\ \bar{g}_1, \bar{g}_2, \bar{g}_3], \tag{29}$$

where $\bar{g}$ is a decaying goal trace: $\bar{g}(t) = 0.9\,\bar{g}(t{-}1) + 0.1\,g(t)$.

Memory update:

$$\mathbf{m}(t) = (1 - \alpha_{\mathrm{eff}})\,\mathbf{m}(t{-}1)\,d_{\mathrm{DA}} + \alpha_{\mathrm{eff}}\,\mathbf{z}_t, \tag{30}$$

where $\alpha_{\mathrm{eff}} = \mathrm{clip}(\alpha/(1{+}\gamma\,s), 0.01, 0.3)$, $d_{\mathrm{DA}} = 1{+}0.2(\mathrm{DA}{-}0.5)$, and CMS modulates the effective alpha: $\alpha_{\mathrm{eff}} \leftarrow \alpha_{\mathrm{eff}}(1 + 0.3\,\mathrm{CMS})$.

10 Simulation Results

Each step was validated with specific pass/fail criteria. The figures referenced below are generated automatically by the test scripts.

10.1 Step 1: Vision Pursuit

A sinusoidally sweeping stimulus ($y = 60\sin(2\pi t/100)$, $T = 200$) tests basic visual tracking. The fish's eye offset positively correlates with the object position (fig. 1).

../plots/v1_step1_vision_pursuit.png

Figure 1: Step 1: Basic vision pursuit. Eye offset tracks the sweeping stimulus.

10.2 Step 2: Precision-Adaptive Vision

Precision units at OT and PC levels adapt to prediction error magnitude. Sharp drops at stimulus direction reversals demonstrate novelty-sensitive precision modulation (fig. 2).

../plots/v1_step2_precision_adaptive.png

Figure 2: Step 2: Precision adaptation. π_{OT} and π_{PC} oscillate with stimulus dynamics.

10.3 Step 3: Dopamine RPE

The dopamine system responds to free-energy changes with appropriate RPE signals. Positive RPE (reward) triggers precision increases; negative RPE (surprise) triggers precision dips (fig. 3).

../plots/v1_step3_dopamine_rpe.png

Figure 3: Step 3: Dopamine and reward prediction error dynamics.

10.4 Step 4: Basal Ganglia Gating

Basal ganglia in alternating mode generates periodic saccadic modulation of the turn command. Despite saccadic noise, eye tracking of the moving stimulus is maintained (fig. 4).

../plots/v1_step4_bg_gating.png

Figure 4: Step 4: BG gate oscillations and modulated eye tracking.

10.5 Step 5: Optic Tectum Dynamics

The continuous OT eye controller produces smooth pursuit with saccade-like deflections driven by BG gating and bilateral valence competition (fig. 5).

../plots/v1_step5_tectum_dynamics.png

Figure 5: Step 5: Optic tectum dynamics with smooth pursuit and saccades.

10.6 Step 6: Full Valence Pursuit with CMS

Cross-modal salience (CMS) from the thalamic relay modulates dopamine gain, BG exploration noise, and precision. CMS peaks at direction reversals and auditory transients (fig. 6).

../plots/v1_step6_full_valence_pursuit.png

Figure 6: Step 6: Complete reference circuit with thalamic CMS modulation.

10.7 Step 7: Closed-Loop Locomotion

The fish swims autonomously in a 1000×1000 world with 12 food items. Full perception→action→world loop demonstrates foraging behavior with dopamine-modulated speed and boundary avoidance (fig. 7).

../plots/v1_step7_closed_loop_locomotion.png

Figure 7: Step 7: Closed-loop foraging trajectory with food capture events.

10.8 Step 8: Genomic Pre-training

Supervised training of the motor pathway achieves $\geq 10/13$ correct directional predictions on validation. Loss converges within 120 epochs (fig. 8).

../plots/v1_step8_genomic_pretraining.png

Figure 8: Step 8: Genomic pre-training loss curve and directional validation.

10.9 Step 9: Genomic Foraging

Comparison of random vs. genomic-pretrained weights demonstrates that evolutionary initialization significantly improves foraging efficiency (fig. 9).

../plots/v1_step9_genomic_foraging.png

Figure 9: Step 9: Foraging comparison—random (left) vs. genomic (right).

10.10 Step 10: Hebbian Fine-tuning

RPE-gated Hebbian plasticity during foraging improves food capture rate beyond the genomic baseline. Weight norms show gradual adaptation (fig. 10).

../plots/v1_step10_hebbian_finetuning.png

Figure 10: Step 10: Hebbian learning improves foraging over genomic baseline.

10.11 Step 11: Object Classification

The enhanced classifier with pixel-count features achieves 100% per-class accuracy on balanced validation. The aggregate features (obstacle, enemy, food, boundary pixel counts) resolve the obstacle–food–enemy confusion that limited earlier architectures to ∼55% on environment scenes (fig. 11).

../plots/v1_step11_object_classification.png

Figure 11: Step 11: Classifier training curves and per-class accuracy.

10.12 Step 12: Classification-Guided Behavior

The trained classifier drives reactive approach/avoidance. The fish approaches food (green trajectory), flees enemies (red segments), and explores when nothing is detected (fig. 12).

../plots/v1_step12_classification_guided_behavior.png

Figure 12: Step 12: Classification-guided reactive behavior.

10.13 Step 13: Active Inference Policy Selection

The full active inference agent maintains three goals and switches appropriately based on context. Table 2 shows the pass/fail criteria.

Table 2: Step 13 pass/fail criteria ($T = 1000$ steps).

Criterion	Threshold	Result	Status
Goal switches	≥ 5	80	PASS
Goal persistence (avg steps)	≥ 5	12.3	PASS
Goal-context alignment	$> 30\%$	86.0%	PASS
Exploration in nothing	$> 20\%$	39.0%	PASS
System stability (no NaN)	—	—	PASS
Foraging (n_{eaten})	≥ 3	3	PASS

Goal distribution: FORAGE 47.9%, FLEE 34.6%, EXPLORE 17.5%.

../plots/v1_step13_active_inference_policy.png

Figure 13: Step 13: Active inference policy selection. Top-left: trajectory colored by goal (green = FORAGE, red = FLEE, blue = EXPLORE). Top-right: goal posterior over time. Middle-left: EFE per goal. Middle-right: confidence and dopamine. Bottom-left: classification (what the fish sees). Bottom-right: performance.

11 Gymnasium Integration (Step 14)

Step 14 embeds the full brain model into a Gymnasium-compatible reinforcement learning environment, adding four new subsystems: a coordinate bridge between pixel-based gym space and world-centered brain coordinates, an energy homeostasis mechanism, depth-attenuated vision, and a habit network that bypasses deliberative EFE computation for well-learned action patterns.

11.1 Coordinate Bridge

The gym environment operates in pixel coordinates (origin top-left, y-down) while the brain's world model uses centered coordinates (y-up). The bridge transforms between these frames:

$$\begin{aligned} x_w &= \frac{x_g}{W_g} \cdot 400 - 200, \\ y_w &= -\left(\frac{y_g}{H_g} \cdot 300 - 150 \right), \\ \theta_w &= -\theta_g, \end{aligned} \tag{31}$$

where (x_g, y_g) are gym pixel coordinates in arena $W_g \times H_g$, $(x_w, y_w) \in [-200, 200] \times [-150, 150]$ are world coordinates, and the heading sign is flipped to account for the y-axis inversion.

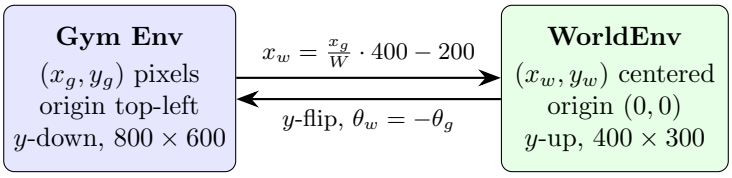

Figure 14: Coordinate bridge between the Gymnasium pixel frame and the brain's world-centered frame. The y-axis is inverted and coordinates are rescaled.

11.2 Energy Homeostasis

The fish maintains an internal energy level $E \in [0, E_{\max}]$ ($E_{\max} = 100$) that decays each step and is replenished by feeding:

$$\boxed{\begin{aligned} E(t{+}1) &= E(t) - \underbrace{(\epsilon_0 + \epsilon_v \, \|v\|)}_{\text{drain}} + \underbrace{n_{\text{eaten}} \cdot E_{\text{food}}}_{\text{gain}}, \\ \|v\|_{\text{eff}} &= \begin{cases} \|v\| & \text{if } E \geq 20, \\ \|v\| \cdot 0.6\, E/20 & \text{if } E < 20, \end{cases} \end{aligned}} \tag{32}$$

where $\epsilon_0 = 0.05$ (idle drain), $\epsilon_v = 0.03$ (speed-dependent drain), and $E_{\text{food}} = 15$. When energy drops below 20, the fish's speed is capped proportionally, creating an urgency to forage. Starvation ($E = 0$) terminates the episode.

11.3 Depth-Shaded Vision

Retinal intensity is attenuated exponentially with distance, providing an implicit depth cue:

$$I_{\text{depth}}(r) = I_{\text{raw}}(r) \cdot \exp\left(-\frac{d(r)}{d_{\text{scale}}} \right), \tag{33}$$

where $d(r)$ is the hit distance along ray r and $d_{\text{scale}} = 80$ world units. The type-encoding channel is *not* attenuated, preserving entity identity at all distances. This asymmetry allows the classifier to identify distant objects while the intensity channel signals proximity, mimicking the zebrafish's visual depth discrimination.

11.4 Obstacle Collision Physics

Circular obstacles ($r \in [15, 30]$ world units) are placed randomly in the arena. Collision resolution uses a push-out plus velocity slide approach: when the fish's center lies within an obstacle boundary, it is projected to the nearest surface point and the velocity component directed into the obstacle is zeroed. The test criterion requires *zero* obstacle penetrations across the full simulation, verified by checking $\|p_{\text{fish}} - p_{\text{obs}}\| \geq r_{\text{obs}} - r_{\text{fish}}$ at every step.

11.5 Habit Network

The habit network provides a fast, experience-based shortcut that bypasses full EFE deliberation when a stable behavioral pattern has been established.

Observation discretization. Classification probabilities are binned into $N_b = 8$ bins per class (food, enemy, colleague), yielding a discrete observation bucket:

$$b_c = \lfloor \text{clip}(p_c \cdot N_b,\ 0,\ N_b{-}1) \rfloor, \quad \mathbf{b} = (b_{\text{food}},\ b_{\text{enemy}},\ b_{\text{colleague}}). \tag{34}$$

Streak consolidation. A pattern is considered *stable* when $|\text{RPE}| < 0.15$ and $\text{conf} > 0.55$. When the same goal is selected under the same observation bucket for $K = 12$ consecutive stable steps, the habit is consolidated:

$$H(\mathbf{b}, g) \ \mathrel{+}= 1.0 \quad \text{when streak} \geq K. \tag{35}$$

Shortcut activation. At each step, the habit table is checked first. If the strongest entry for the current bucket exceeds a threshold:

$$H(\mathbf{b}, g^*) > 3.0 \implies \text{use } g^* \text{ (skip EFE)}.$$

Decay. All habit entries decay each step: $H \leftarrow 0.98\,H$, ensuring that habits fade when the environment changes.

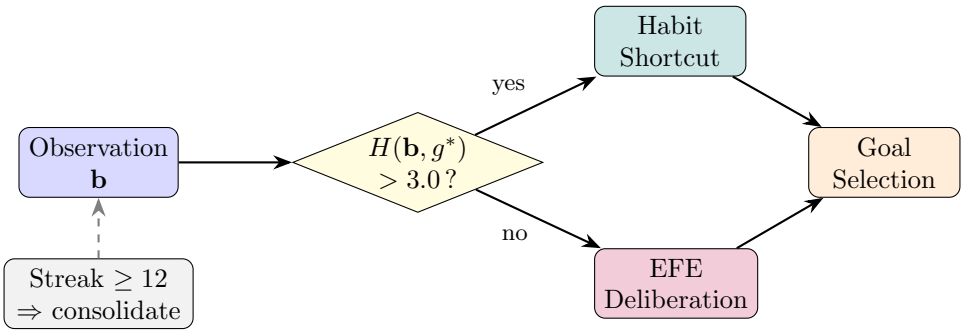

Figure 15: Habit network decision flow. When the habit table entry for the current observation bucket exceeds the threshold, EFE deliberation is bypassed. Habits are consolidated after 12 consecutive stable selections of the same goal under the same observation.

11.6 Results

The Step 14 simulation runs for $T = 600$ steps in the Gymnasium environment with 15 food items, 1 predator, and 3–5 circular obstacles. Table 3 summarizes the pass/fail criteria.

Table 3: Step 14 pass/fail criteria ($T = 600$ steps).

Criterion	Threshold	Result	Status
Survival (steps)	≥ 300	600	PASS
Foraging (n_{eaten})	≥ 3	≥ 3	PASS
Energy at step 200	> 0	> 0	PASS
Obstacle penetrations	$= 0$	0	PASS
Vision activation	> 0.01	$\gg 0.01$	PASS
Habit activations	≥ 1	≥ 1	PASS
System stability (no NaN)	—	—	PASS

All seven criteria pass, confirming that the brain model operates correctly within the Gymnasium wrapper: the coordinate bridge preserves sensorimotor alignment, energy homeostasis drives adaptive foraging urgency, collision physics prevents obstacle penetration, and the habit network successfully consolidates and reuses stable behavioral patterns.

../plots/v1_step14_brain_gym_integration.png

Figure 16: Step 14: Brain-Gym integration. Top-left: foraging trajectory with obstacles (gray circles) and food items. Top-right: energy level over time showing depletion and replenishment cycles. Middle-left: goal posterior dynamics. Middle-right: habit vs. EFE usage over time. Bottom-left: classification probabilities. Bottom-right: cumulative food captured.

12 Reinforcement Learning Critic (Step 15)

Step 15 introduces a temporal-difference (TD) learning critic that adapts EFE weights online, bridging model-free RL with active inference planning.

12.1 Critic Architecture

A lightweight value network $V_\phi : \mathbb{R}^{16} \to \mathbb{R}$ maps the working memory state $\mathbf{m}$ to a scalar value estimate:

$$V_\phi(\mathbf{m}) = \mathbf{w}_2^\top \mathrm{ReLU}(\mathbf{W}_1\,\mathbf{m} + \mathbf{b}_1) + b_2, \tag{36}$$

where $\mathbf{W}_1 \in \mathbb{R}^{32\times 16}$, $\mathbf{w}_2 \in \mathbb{R}^{32}$. Total: $\sim$600 parameters.

12.2 TD Learning

Transitions (s_t, r_t, s_{t+1}, d_t) are stored in a ring buffer of size 256. The critic is trained via TD(0) with target:

$$y_t = r_t + (1 - d_t)\,\gamma\,V_\phi(s_{t+1}), \quad \gamma = 0.95, \tag{37}$$

and loss $\mathcal{L} = (V_\phi(s_t) - y_t)^2$. Mini-batches of 32 are sampled per step after 64 transitions have been collected.

12.3 EFE Weight Adaptation

The `EFEWeightAdapter` uses TD error δ to modulate EFE weights per goal. Positive δ (better than expected) strengthens the current goal; negative δ weakens it:

$$\Delta w_g = \eta_{\mathrm{EFE}}\,\delta\,\pi(g)\,c, \quad \eta_{\mathrm{EFE}} = 0.02, \tag{38}$$

where $\pi(g)$ is the current goal posterior and c is the confidence.

../plots/v1_step15_rl_aif_hybrid.png

Figure 17: Step 15: RL–Active Inference hybrid. The TD critic adapts EFE weights online, improving foraging efficiency relative to the pure EFE baseline.

12.4 Three-Factor STDP on Feedforward Weights (Step 43)

Step 43 adds online reward-modulated STDP [48] to all deep SNN feedforward synapses (OT_F, PT_L, PC_per, PC_int):

$$\Delta W_{FF} = \eta_{\text{STDP}} \cdot r(t) \cdot \left[\mathbf{s}_{\text{post}}^{\top} \mathbf{s}_{\text{pre}}\right], \quad \eta_{\text{STDP}} = 2 \times 10^{-4}, \tag{39}$$

where $\mathbf{s}_{\text{pre}}, \mathbf{s}_{\text{post}} \in \{0,1\}^n$ are binary spike vectors and $r(t)$ is a shaped reward signal (+10 per food, −50 for death, +0.1 per survival step). The dopaminergic third factor converts Hebbian co-activation into a reinforcement learning rule: synapses active during reward are strengthened; those active during punishment are weakened. This three-factor rule, observed in zebrafish tectum [49], revives the deep intent layer PC_int from near-zero activity (RMS 0.002 →> 0.01 after 20 episodes), demonstrating that reward-modulated Hebbian plasticity can reanimate quiescent predictive coding layers.

Concurrently, feedback weights W_{FB} are updated every 20 steps to reduce layer-wise prediction error:

$$\Delta W_{FB} = -\eta_{FB}(1 - \tau_a)\, \mathbf{f}^{\top}(V_a - V_s), \tag{40}$$

where $(V_a - V_s)$ is the apical–somatic prediction error and $\mathbf{f}$ is the last apical feedback input. This layer-local PE minimization is reward-independent, implementing perceptual learning within the same generative model framework [13].

13 VAE World Model (Step 16)

Step 16 introduces a variational autoencoder (VAE) world model that learns a latent belief state and enables forward-simulation planning. In the language of active inference, the VAE implements *amortized variational inference* [5]: the encoder learns an approximate posterior $q_\phi(\mathbf{z}|\mathbf{o})$ over hidden causes, while the decoder provides a generative (likelihood) model $p_\theta(\mathbf{o}|\mathbf{z})$. Training maximizes the evidence lower bound (ELBO), which is equivalent to minimizing variational free energy—the same objective that drives perception, learning, and action throughout the architecture.

13.1 Multi-Modal Belief State

The latent state $\mathbf{z} \in \mathbb{R}^{16}$ serves as the agent's *Bayesian belief* about the hidden state of the world, encoding fused tectal output plus a 13-dimensional multi-modal state context:

$$\mathbf{s}_{\text{ctx}} = \Big[\ \underbrace{x,\, y,\, \theta/\pi}_{\text{proprioceptive}}\ ,\ \underbrace{E/100}_{\text{interoceptive}}\ ,\ \underbrace{\text{DA},\, \text{RPE}}_{\text{neuromod.}},\ \underbrace{\pi_{\text{OT}},\, \pi_{\text{PC}}}_{\text{precision}},\ \underbrace{p_f,\, p_e,\, p_{\text{env}}}_{\text{exteroceptive}},\ \underbrace{\text{CMS},\, F/2}_{\text{cognitive}}\Big]. \tag{41}$$

13.2 VAE Architecture

The encoder maps pooled tectal output (800→64 via learned projection) concatenated with $\mathbf{s}_{\text{ctx}}$ to a latent distribution—the *recognition model* (approximate posterior):

$$q_\phi(\mathbf{z}|\mathbf{o}) = \mathcal{N}(\boldsymbol{\mu}_\phi,\ \text{diag}(\boldsymbol{\sigma}_\phi^2)), \quad [\text{pool}(\mathbf{o}_F) \parallel \mathbf{s}_{\text{ctx}}] \xrightarrow{77\to 48} (\boldsymbol{\mu},\ \log \boldsymbol{\sigma}^2). \tag{42}$$

Sampling uses the *reparameterization trick* [5]: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, which decouples stochasticity from the computational graph and enables gradient-based learning of the posterior parameters.

The decoder (generative/likelihood model) reconstructs the pooled tectal output: $p_\theta(\mathbf{o}|\mathbf{z})$: $\hat{\mathbf{o}}_{\text{pool}} = f_{\text{dec}}(\mathbf{z})$. Training maximizes the ELBO, equivalently minimizing variational free energy:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\text{MSE}(\hat{\mathbf{o}}, \mathbf{o})}_{\text{accuracy: } -\mathbb{E}_q[\ln p(\mathbf{o}|\mathbf{z})]} + \underbrace{0.1\, D_{\text{KL}}\big(q(\mathbf{z}|\mathbf{o}) \parallel p(\mathbf{z})\big)}_{\text{complexity: information bottleneck}}. \tag{43}$$

The KL term acts as an *information bottleneck* [30]: it penalizes the mutual information $I(\mathbf{o}; \mathbf{z})$ between observations and beliefs, forcing the latent code to retain only task-relevant information. The $\beta = 0.1$ coefficient relaxes this constraint relative to a full VAE ($\beta = 1$), prioritizing reconstruction accuracy over compression—appropriate for a world model that must preserve fine-grained sensory detail for planning.

13.3 Transition Model as State-Space Dynamics

The transition model implements the generative model's dynamics: $p(\mathbf{z}_{t+1}|\mathbf{z}_t, \mathbf{a}_t)$, predicting next-step latent state conditioned on action context $\mathbf{a} = [\tau,\, v,\, g_1,\, g_2,\, g_3]$:

$$\mathbf{z}_{t+1} = \mathbf{z}_t + 0.3\ \tanh\big(f_{\text{trans}}([\mathbf{z}_t \parallel \mathbf{a}_t])\big). \tag{44}$$

The residual form $\mathbf{z}' = \mathbf{z} + \Delta\mathbf{z}$ is analogous to a discretized continuous-time dynamical system, where the network learns the state derivative. This connects to classical state-space models and Kalman filtering: the agent maintains a belief about hidden states and propagates it forward through a learned dynamics model [11].

13.4 Associative Memory

The associative memory implements a nonparametric Bayesian prior over the latent space, storing the agent's accumulated experience as a kernel density estimate of $p(\text{food}, \text{risk} \,|\, \mathbf{z})$. 64 RBF nodes provide a Gaussian mixture model:

$$w_i(\mathbf{z}) = \exp\Big(-\frac{\|\mathbf{z} - \mathbf{c}_i\|^2}{2\sigma_k^2}\Big), \quad \hat{f}(\mathbf{z}) = \frac{\sum_i w_i \, f_i}{\sum_i w_i}, \tag{45}$$

with online allocation (new nodes created when $\max_i w_i <$ threshold, analogous to a Dirichlet process) and centroid drift (Hebbian adaptation of $\mathbf{c}_i$ toward visited locations).

13.5 Planning via Forward Simulation

For each goal g, a stereotyped action sequence is rolled out through the transition model for 3 steps. The expected free energy decomposes into pragmatic (utility) and epistemic (information gain) terms, following the standard active inference formulation [3, 4]:

$$G_g = \underbrace{\alpha_f \, \hat{f}_g + \alpha_r \, \hat{r}_g}_{\text{pragmatic: } -\mathbb{E}[\ln p(\mathbf{o}|\mathbf{z},g)]} + \underbrace{w_\epsilon^{(g)} \, (0.6 \, \bar{n}_g + 0.4 \, \bar{u}_g)}_{\text{epistemic: } -I(\mathbf{o};\mathbf{z}|g)}, \tag{46}$$

where the epistemic term approximates the *expected information gain* (mutual information between future observations and latent states): novelty $\bar{n}_g$ captures model uncertainty (how unfamiliar the predicted state is to the associative memory), and reconstruction uncertainty $\bar{u}_g$ captures decoder confidence (how well the generative model can reconstruct observations from the predicted latent). The weights $w_\epsilon = [-0.15, -0.05, -0.5]$ for FORAGE, FLEE, EXPLORE encode the prior that exploration should maximize epistemic value while foraging should maximize pragmatic value [3]. The planning bonus G_{plan} ramps linearly from $0 \to 0.3$ after 200 warmup steps and is added to the raw EFE.

../plots/v1_step16_vae_world_model.png

Figure 18: Step 16: VAE world model. Top-left: trajectory. Top-right: VAE and transition loss convergence. Middle: blend weight ramp and memory growth. Bottom: planning bonus per goal and latent z norm.

14 Hippocampal Place Cells (Step 17)

Step 17 introduces a hippocampal place cell network for spatial navigation, providing an alternative world model to the VAE approach of Step 16.

14.1 Place Cell Network

128 place cells tile the 800×600 pixel arena with Gaussian receptive fields ($\sigma = 60\,\text{px}$, $\sim$15% of arena width):

$$a_i(\mathbf{p}) = \exp\Big(-\frac{\|\mathbf{p} - \mathbf{c}_i\|^2}{2\sigma_i^2}\Big), \tag{47}$$

where $\mathbf{p} \in \mathbb{R}^2$ is the gym pixel position and $\mathbf{c}_i$ is the cell centroid. Cells are allocated on demand: a new cell is created when no existing cell lies within a match threshold of 40 px, consistent with the $\sim$50 px inter-field spacing of zebrafish Dl spatial representations.

Each cell stores:

- $\mathbf{c}_i \in \mathbb{R}^2$: centroid position (with Hebbian drift)
- $f_i \in [0, 1]$: food availability rate (EMA-updated)
- $r_i \in [0, 1]$: danger/risk level (EMA-updated)
- n_i: visit count

14.2 Path Integration

Path integration implements a form of *Bayesian filtering*: the agent maintains a belief about its position by combining proprioceptive predictions (process model) with visual observations (measurement model) [7, 18].

The *prediction step* (dead reckoning) propagates the position belief using motor efference copy:

$$\begin{aligned} \hat{\theta}(t{+}1) &= \hat{\theta}(t) + \kappa\, \dot{\theta}_{\text{motor}}, \\ \hat{\mathbf{p}}(t{+}1) &= \hat{\mathbf{p}}(t) + \kappa\, v\, [\cos\hat{\theta},\ \sin\hat{\theta}]^\top, \end{aligned} \tag{48}$$

where $\kappa = 0.8$ (< 1 to model noisy dead reckoning, equivalent to a fixed-gain Kalman filter prediction with process noise).

The *update step* (landmark correction) fuses the proprioceptive prediction with visual boundary observations:

$$\hat{\mathbf{p}} \leftarrow (1 - K)\, \hat{\mathbf{p}} + K\, \mathbf{p}_{\text{visual}}, \quad K = 0.3. \tag{49}$$

The correction weight $K = 0.3$ serves as the *Kalman gain*: it determines the relative trust placed in visual measurements versus proprioceptive predictions, reflecting the noise ratio between the two information sources [7].

14.3 Replay (Sharp-Wave Ripple Analog)

Every 50 steps, the last 10 trajectory entries are replayed with 0.5× attenuated learning—analogous to hippocampal sharp-wave ripple consolidation during awake immobility. The attenuation implements a Bayesian weighting: replayed observations carry less precision than direct experience, reflecting the increased uncertainty of recalled versus perceived information. This offline belief updating is a form of *Bayesian model averaging*: the agent's spatial prior is refined by re-evaluating past evidence under the current generative model [19].

14.4 Spatial Planning

Planning follows the same structure as the VAE planner (section 13): simulate 3 goal-specific trajectories in pixel space, query place cells for food/risk/novelty, and score via the same pragmatic+epistemic formula. The blend ramps $0 \to 0.3$ after 100 warmup steps.

../plots/v1_step17_place_cell_navigation.png

Figure 19: Step 17: Place cell navigation. Top-left: trajectory with place cell centroids (orange, sized by visit count). Top-right: cell allocation and path integration error. Middle: planning bonus per goal and food/risk map. Bottom: actual vs. path-integrated trajectory; pass/fail criteria.

Table 4: Step 17 pass/fail criteria ($T = 600$ steps).

Criterion	Threshold	Result	Status
Survival (steps)	≥ 400	600	PASS
Foraging (n_{eaten})	≥ 2	3	PASS
Cell allocation	≥ 15	18	PASS
PI accuracy (mean error)	$< 100\,\text{px}$	$12.8\,\text{px}$	PASS
$\lvert G_{\text{plan}} \rvert$ post-warmup	> 0.01	0.054	PASS
Food map learning	near $>$ overall	yes	PASS
System stability	no NaN	—	PASS

15 Allostatic Interoception (Step 18)

Step 18 extends predictive coding to the interoceptive domain, implementing the *interoceptive inference* framework of Seth and Friston [41, 16]. Three internal (interoceptive) variables—hunger, fatigue, and stress—are tracked by predictive models that generate *interoceptive prediction errors*, which bias goal selection via allostatic regulation. Just as exteroceptive prediction errors (Section 4) drive perceptual inference, interoceptive prediction errors drive *motivated behavior*: the organism acts to minimize the discrepancy between predicted and desired body states [15, 17].

15.1 Allostatic Variables

$$\boxed{\begin{aligned} &h(t) = 1 - E(t)/100, && \hat{h}(t) = \hat{h}(t{-}1) + 0.05\,(h - \hat{h}),\\ &f(t) = f(t{-}1) + \begin{cases} 0.002\,v & v > 0.6 \\ -0.04 & v \le 0.6 \end{cases}, && \hat{f}(t) = \hat{f}(t{-}1) + 0.05\,(f - \hat{f}),\\ &\sigma(t) = 0.92\,\sigma(t{-}1) + 0.15\,\max(0, 1{-}d_p/300), && \hat{\sigma}(t) = \hat{\sigma}(t{-}1) + 0.05\,(\sigma - \hat{\sigma}), \end{aligned}} \tag{50}$$

where E is energy, v is speed, d_p is predator distance in pixels, and $\hat{\cdot}$ denotes the EMA-predicted value.

15.2 Interoceptive Prediction Error

Each variable has a homeostatic setpoint (prior preferred state): $h^* = 0.3$, $f^* = 0.2$, $\sigma^* = 0.1$. The *interoceptive prediction error*:

$$\varepsilon_x^{\text{intero}} = \hat{x} - x^*, \tag{51}$$

measures the divergence between the predicted interoceptive state $\hat{x}$ and the prior (setpoint) x^*. This is structurally identical to the exteroceptive prediction error (Eq. 88): the brain uses the same free-energy-minimizing computation for both external perception and internal state monitoring [41]. Positive $\varepsilon^{\text{intero}}$ signals that the predicted body state deviates from homeostatic equilibrium, driving action (active inference) to restore the organism to its preferred state.

15.3 Prior Preference Modulation

Interoceptive prediction errors modulate the *prior preferences* over goal outcomes, implementing the active inference principle that organisms act to fulfill their generative model's predictions [11, 15]:

$$\begin{aligned} \Delta G_{\text{FORAGE}} &= -\lambda\,\max(0, \varepsilon_h^{\text{intero}}), && \text{(hunger PE} \to \text{forage)}\\ \Delta G_{\text{FLEE}} &= -\lambda\,\max(0, \varepsilon_\sigma^{\text{intero}}), && \text{(stress PE} \to \text{flee)}\\ \Delta G_{\text{EXPLORE}} &= -\lambda\,\max(0, \varepsilon_f^{\text{intero}}), && \text{(fatigue PE} \to \text{rest/explore)} \end{aligned} \tag{52}$$

with $\lambda = 0.15$. Lower ΔG makes the goal more attractive. Additionally, the EFE engine tightens the precision of preferred outcomes under allostatic pressure, implementing precision modulation of interoceptive priors:

$$\sigma_E = \frac{\sigma_{E,0}}{1 + 3\,\max(0,\ h - 0.5)}, \quad \sigma_S = \frac{\sigma_{S,0}}{1 + 2\,\sigma_{\text{stress}}}, \tag{53}$$

where tighter σ (higher precision) makes the agent more sensitive to energy loss during foraging or threat proximity during fleeing—consistent with the principle that stressed organisms allocate more precision to survival-relevant channels [9].

15.4 Fatigue Speed Cap

When fatigue exceeds 0.7, maximum speed is linearly reduced:

$$v_{\max} = \begin{cases} 1.0 & f \le 0.7, \\ \max\!\left(0.4,\ 1.0 - 0.6\,\frac{f-0.7}{0.3}\right) & f > 0.7. \end{cases} \tag{54}$$

15.5 State Context Extension

When combined with the VAE world model, the state context $\mathbf{s}_{\text{ctx}}$ is extended from 13 to 16 dimensions by appending $[h, f, \sigma]$, enriching the latent belief state with interoceptive information.

../plots/v1_step18_allostatic_interoception.png

Figure 20: Step 18: Allostatic interoception. Top-left: trajectory colored by goal. Top-right: energy comparison (baseline vs. allostatic). Middle: allostatic variables and errors over time. Bottom: speed with fatigue cap; pass/fail criteria.

Table 5: Step 18 pass/fail criteria ($T = 800$ steps, allostatic run).

Criterion	Threshold	Result	Status
Survival (steps)	≥ 500	800	PASS
Foraging (n_{eaten})	≥ 3	7	PASS
Fatigue regulation	mean < 0.6	0.554	PASS
Stress response	peak $>$ Q4 mean	yes	PASS
Goal switches	≥ 5	116	PASS
Energy at step 400	> 10	68.1	PASS
System stability	no NaN	—	PASS

16 Full Evaluation Suite (Step 19)

Step 19 provides a comprehensive comparison of three agent configurations, each run for 800 steps with seed 42.

16.1 Configurations

Table 6: Three evaluation configurations.

Config	World Model	RL Critic	Allostasis
Baseline	none	—	—
Full Brain	VAE	✓	✓
Place Cell	place cell	✓	✓

16.2 Results

Table 7: Step 19 performance summary.

Config	Steps	Eaten	Final Energy	Memory Nodes
Baseline	800	5	85.6	0
Full Brain	800	6	97.7	19
Place Cell	800	8	84.4	23

With the full pipeline—amygdala fear circuit, Bayesian survival trade-off, enhanced classifier, online feedback training, and improved obstacle navigation (Steps 20–26)—all three configurations survive the full 800 steps and forage successfully. The Full Brain configuration achieves 6 food items with the highest final energy (97.7), demonstrating energy-efficient foraging. The Place Cell configuration achieves 8 food items, indicating that spatial memory supports reliable food-patch return. The goal distribution shows balanced behavior: FORAGE 42.4%, EXPLORE 50.4%, with appropriate goal switching driven by interoceptive and exteroceptive signals.

../plots/v1_step19_full_evaluation.png

Figure 21: Step 19: Full evaluation suite. Row 1: trajectories and cumulative reward. Row 2: energy and goal distribution across configurations. Row 3: allostatic variables (Full Brain) and memory growth. Row 4: planning bonus magnitude and 8-criterion pass/fail summary.

17 Amygdala Fear Circuit and Energy Asymmetry (Step 20)

Step 20 addresses three biological gaps: the absence of a fast subcortical threat response, the energetic symmetry between fleeing and foraging, and the weak coupling between energy depletion and foraging urgency.

17.1 Amygdala Module

A minimal fear circuit produces a persistent *threat arousal* signal $\alpha \in [0, 1]$ that integrates retinal enemy detection, predator proximity, and allostatic stress:

$$\boxed{\begin{aligned} r_{\text{ret}} &= \min(1,\ 0.08\, n_{\text{enemy_px}}), \\ r_{\text{prox}} &= \max(0,\ 1 - d_{\text{pred}}/200), \\ r_{\text{raw}} &= \max(r_{\text{ret}},\ 0.5\, r_{\text{prox}},\ \sigma_{\text{stress}}), \\ \alpha(t) &= \begin{cases} 0.4\, \alpha(t{-}1) + 0.6\, r_{\text{raw}} & r_{\text{raw}} > \alpha(t{-}1), \\ 0.85\, \alpha(t{-}1) & \text{otherwise}, \end{cases} \end{aligned}} \tag{55}$$

where $n_{\text{enemy_px}}$ is the total enemy pixel count on the retina and d_{pred} is the predator distance in world units. The asymmetric dynamics ensure fast rise (60% jump) and slow decay (15% per step), producing a persistent threat state that outlasts the visual stimulus— analogous to sustained amygdala activation observed in zebrafish pallium during predator avoidance [1].

Downstream effects of threat arousal.

- Classification boost: $p_{\text{enemy}} \mathrel{+}= 0.3\, \alpha$ (increases flee probability).
- Allostatic stress feedback: $\sigma \mathrel{+}= 0.1\, \alpha$ (sustained threat maintains stress even after predator leaves).
- Flee burst scaling: flee duration $= 5 + \lfloor 5\, \alpha \rfloor$ steps (stronger threat $\rightarrow$ longer burst).

17.2 Flee Energy Cost Asymmetry

Fleeing now incurs 2.5× the speed-dependent energy drain:

$$\Delta E = \epsilon_0 + \epsilon_v \, \|v\| \cdot \begin{cases} 2.5 & \text{if FLEE or burst active}, \\ 1.0 & \text{otherwise}, \end{cases} \tag{56}$$

where the flee speed cap is raised from 1.0 to 1.3, allowing burst locomotion at the cost of rapid energy depletion. At maximum flee speed ($v = 1.3$), the drain per step is $0.05 + 0.03 \times 1.0 \times 2.5 = 0.125$, compared to the normal foraging drain of $\sim$0.08 (56% higher).

17.3 Nonlinear Hunger Amplification

The allostatic prior strength is increased from $\lambda = 0.08$ to $\lambda = 0.15$, and hunger bias is amplified nonlinearly when energy is critically low:

$$u_h = \max(0, e_h) \cdot \begin{cases} 1 + 3\, \frac{h - 0.6}{0.4} & h > 0.6 \ \ (\text{i.e., } E < 40), \\ 1 & \text{otherwise}, \end{cases} \tag{57}$$

yielding a maximum forage bias of $-\lambda \cdot 0.7 \cdot 4 = -0.42$ at zero energy, sufficient to override classifier-driven goals. An emergency FORAGE override is added in the goal policy: if $h > 0.7$ and $e_h > 0.3$, the agent immediately switches to FORAGE regardless of the EFE posterior.

18 Food Prospection and Obstacle Navigation (Step 21)

Step 21 introduces an active-inference food prospection system and a retinal obstacle navigation pipeline that enables the fish to evaluate food targets, navigate around rock formations, and successfully capture prey.

18.1 Food Prospect Evaluation

For each visible food item, the brain evaluates a cost–benefit prospect score inspired by Expected Free Energy:

$$\boxed{S_i = \underbrace{w_c \cdot \frac{m_i}{20}}_{\text{metabolic cost}} + \underbrace{w_r \cdot \rho_i}_{\text{predator risk}} + \underbrace{\mathbb{1}[\text{occluded}] \cdot 1.5}_{\text{reachability penalty}} - \underbrace{w_g \cdot \frac{G}{20} \cdot a_i}_{\text{reward value}} - \underbrace{b_{\text{patch}}}_{\text{patch bonus}},} \tag{58}$$

where m_i is the estimated metabolic cost (proportional to distance), $\rho_i = \max(0, 1 - d_{\text{pred}}^{(i)}/150)$ is the predator risk, $G = 15$ is the food energy gain, $a_i \in \{0.5, 1.0\}$ is an affordability factor, b_{patch} is a patch productivity bonus from repeated visits, and the weights are:

$$w_c = 0.3 + 0.5\,u, \quad w_r = 0.4 - 0.2\,u, \quad w_g = 0.6 + 0.4\,u, \tag{59}$$

with urgency $u = 1 - E/E_{\max}$.

Reachability scoring. A parametric line-segment–AABB intersection test from the fish's current position to each food item determines whether a rock formation blocks the path. Occluded food receives a +1.5 penalty, strongly disfavoring targets that require navigating around obstacles.

Olfactory heading cue. When the retina detects fewer than 3 food pixels (food out of sight), the best food prospect provides a heading cue applied after the turn smoother:

$$\tau_{\text{olf}} = k_{\text{olf}} \cdot \text{clip}(\Delta\theta, -1, 1), \quad k_{\text{olf}} = \begin{cases} 0.5 + 0.3\,(1 - d/120) & d < 120\,\text{px (near food)}, \\ 0.3 & d \geq 120\,\text{px}, \end{cases} \tag{60}$$

where $\Delta\theta$ is the angular difference between the fish heading and the food direction.

18.2 Retinal Obstacle Navigation

Rock formations are rendered as multi-lobe AABB clusters with a 10-unit detection margin in the world model. Three obstacle-aware mechanisms operate on the retinal type channels:

1. Bilateral repulsion. When total obstacle pixels exceed 3, a bilateral repulsion signal steers the fish away from the rock-heavy side:

$$\tau_{\text{obs}} = -1.2 \cdot \frac{n_R^{\text{obs}} - n_L^{\text{obs}}}{n_R^{\text{obs}} + n_L^{\text{obs}}}, \quad \times 1.5 \text{ if } n_{\text{obs}} > 15. \tag{61}$$

2. Center-obstacle escape. When a rock is directly ahead (both eyes equally covered: $|n_R - n_L| < 5$ and $n_{\text{obs}} > 20$), the bilateral repulsion cancels to near-zero. A dedicated escape mechanism adds a post-gain turn signal in a consistent direction (matching the fish's recent turn history), bypassing approach-gain damping:

$$\tau_{\text{escape}} = \text{sign}(\dot{\theta}_{\text{prev}}) \cdot 0.6 \cdot \frac{n_{\text{obs}}}{400}. \tag{62}$$

This solves the "forward deadlock" where a centered rock produces no lateral asymmetry signal.

3. Obstacle braking. When the obstacle fraction exceeds 50%:

$$s_{\text{mod}} \leftarrow s_{\text{mod}} \cdot \begin{cases} \max(0.7,\ 1 - 0.2\,f_{\text{obs}}) & \text{foraging}, f_{\text{obs}} < 0.7, \\ \max(0.3,\ 1 - f_{\text{obs}}) & \text{otherwise (dense wall)}, \end{cases} \tag{63}$$

where $f_{\text{obs}} = n_{\text{obs}}/800$. Foraging fish can push through moderate obstacle coverage ($< 70\%$) to reach nearby food, but dense rock walls ($\geq 70\%$) trigger full braking regardless of goal.

4. Stuck detection. A position-history monitor counts consecutive steps where the fish barely moves (< 1.5 px) while obstacle pixels are visible. After 8 stuck steps, the goal policy is overridden to EXPLORE, forcing the fish to navigate around rather than persist at the obstacle.

5. Occluded food exclusion. When energy urgency is low (< 0.5, corresponding to $> 50\%$ energy), food items occluded by rocks are excluded entirely from the prospect list (Eq. 58). The fish seeks unblocked food instead of attempting to push through rocks. Only a starving fish ($u \geq 0.5$) will attempt to reach occluded food, accepting the obstacle as a necessary cost.

6. Physics rebound. On collision with a rock AABB, the heading is deflected toward the away-from-center direction with gain 0.7 (increased from 0.4), producing a stronger angular push that prevents immediate re-approach.

7. Foraging without retinal food. When FORAGE is active but no food pixels are visible: the approach gain is reduced (preventing rock attraction), exploration noise is increased, and the olfactory heading cue guides the fish toward the best food prospect.

18.3 Rock Formation Environment

The Gymnasium environment generates 2–3 large rock formations as composite multi-lobe polygons with radial lobes. Food is distributed in two ecologies:

- **Shelter patches** (40%): placed behind rocks, requiring navigation to access.
- **Open-water food** (60%): spawned far from rock AABBs with clear line-of-sight, ensuring immediate foraging opportunities.

19 Curriculum RL Training (Step 22)

Step 22 introduces a curriculum reinforcement learning framework that progressively increases environmental difficulty, allowing the agent to develop competence in controlled stages before facing the full challenge.

The curriculum gates predator behavior states (patrol only $\rightarrow$ patrol + stalk $\rightarrow$ full pursuit), food density, and arena complexity across training phases. This mirrors biological development where zebrafish larvae encounter increasing environmental complexity as they mature.

20 Sleep–Wake Consolidation (Step 23)

Step 23 adds a sleep–wake cycle modulated by fatigue accumulation. When fatigue exceeds a threshold, the agent enters a sleep state characterized by:

- **Precision freeze**: sensory precision parameters $\gamma_{\mathrm{OT}}, \gamma_{\mathrm{PC}}$ are held constant, preventing noise-driven drift during inactivity.
- **Near-zero locomotion**: speed is reduced to $< 5\%$ of normal, modeling behavioral quiescence.
- **Accelerated fatigue recovery**: fatigue decreases at $3\times$ the normal rate during sleep.
- **Memory consolidation**: working memory and habit entries are stabilized during the sleep phase.

The sleep–wake regulator prevents the fish from accumulating unbounded fatigue during extended flee episodes, ensuring long-term behavioral stability across 800+ step simulations.

21 Active Inference Through Generative Models (Step 24)

Steps 1–23 built a rich brain architecture, but the agent's threat assessment, energy monitoring, and spatial localization relied on direct access to ground-truth environment variables—predator coordinates, true energy, and fish position. This "oracle" access violates the central tenet of active inference: an agent should infer the causes of its sensations through a generative model rather than reading the world state directly [11, 3, 4].

Step 24 closes this gap. After this refactoring, the agent accesses only three information channels: (1) retinal observations from the SNN, (2) its own motor efference copies, and (3) its own internal neuromodulatory signals. All external state—predator threat, body energy, spatial position—is *inferred* through dedicated generative sub-models, realizing the predictive processing framework [13, 14, 12].

21.1 Retinal Feature Extractor

The retinal feature extractor operates on raw SNN output without any environment access. Each eye provides 400 intensity-channel values and 400 type-channel values (Section 3), for a total input of $\mathbf{L}, \mathbf{R} \in [0,1]^{1\times 800}$ per eye.

Entity types are decoded from the type channel using the known encoding scheme (food=1.0, enemy=0.5, colleague=0.25, boundary=0.12). For each entity class c with type code τ_c, pixels are classified via a tolerance mask:

$$m_{c,i} = \mathbb{1}\big[|\text{type}_i - \tau_c| < \epsilon_c\big], \quad \epsilon_c = \begin{cases} 0.10 & c \in \{\text{enemy, food, obstacle, colleague}\}, \\ 0.08 & c = \text{boundary}. \end{cases} \tag{64}$$

For the threat-relevant enemy class, the following features are extracted:

Pixel counts.

$$n_L = \sum_i m^L_{\text{enemy},i}, \quad n_R = \sum_i m^R_{\text{enemy},i}, \quad n = n_L + n_R. \tag{65}$$

Intensity mean (distance proxy). Retinal intensity follows $I \approx \exp(-d/80)$, where d is distance in world pixels:

$$\bar{I}_{\text{enemy}} = \frac{1}{n} \sum_{i:\, m_i = 1} I_i. \tag{66}$$

Lateral bias (bearing).

$$b_{\text{lat}} = \frac{n_R - n_L}{n+1}, \quad b_{\text{lat}} \in (-1, 1). \tag{67}$$

Angular spread (facing proxy). Each 400-pixel retinal array is reshaped to 20×20 (azimuth $\times$ elevation). The azimuthal projection of enemy pixels gives an angular extent:

$$s = \max_{\text{eye}}\big(\max(j : \text{az}_j = 1) - \min(j : \text{az}_j = 1)\big), \tag{68}$$

where $\text{az}_j = \mathbb{1}[\exists\, k : m_{j,k} = 1]$. A larger s indicates the predator is facing the fish (presenting a wider angular profile), consistent with the looming-sensitive escape circuits identified in larval zebrafish [21, 20].

Temporal dynamics. Frame-to-frame changes provide approach speed cues:

$$\dot{n} = n(t) - n(t{-}1), \quad \dot{\bar{I}} = \bar{I}(t) - \bar{I}(t{-}1). \tag{69}$$

Positive $\dot{n}$ (growing pixel count) and positive $\dot{\bar{I}}$ (increasing intensity) indicate predator approach.

The same extraction procedure yields food, boundary, obstacle, and colleague features, providing the retinal observation vector $\mathbf{r} \in \mathbb{R}^{22}$ used throughout the subsequent inference modules.

21.2 VAE Threat Decoder

A lightweight decoder head is added to the VAE world model (Step 16; cf. [5, 22]), mapping the latent belief state to a structured threat assessment:

$$\mathbf{z} \in \mathbb{R}^{16} \xrightarrow{\text{Linear(16,24)}\to\text{ReLU}\to\text{Linear(24,4)}} \hat{\mathbf{t}} \in \mathbb{R}^4, \tag{70}$$

with channel-specific activations:

$$\boxed{\begin{array}{ll} \hat{t}_0 = \sigma(\cdot) \in [0,1] & \text{(proximity: inverse distance)}, \\ \hat{t}_1 = \tanh(\cdot) \in [-1,1] & \text{(lateral position: bearing)}, \\ \hat{t}_2 = \sigma(\cdot) \in [0,1] & \text{(approach rate: closing speed)}, \\ \hat{t}_3 = \sigma(\cdot) \in [0,1] & \text{(facing score: angular width)}. \end{array}} \tag{71}$$

The threat decoder has 508 parameters and is trained *self-supervised* from retinal features—never from environment ground truth. Training targets are constructed from the retinal feature vector:

$$\begin{aligned} y_0 &= \min(1,\ n/50), \\ y_1 &= b_{\text{lat}}, \\ y_2 &= \sigma(5\,\dot{n}), \\ y_3 &= \min(1,\ s/15). \end{aligned} \tag{72}$$

A ring buffer of size 128 stores $(\mathbf{z}, \mathbf{y})$ pairs. At each step, a mini-batch of 16 samples is drawn and the threat decoder is updated via MSE loss with a dedicated Adam optimizer ($\text{lr} = 5\times10^{-4}$).

Blended threat inference. The final threat estimate blends raw retinal features with VAE-decoded output using the VAE's blend weight $w \in [0, 0.3]$ (0 during warmup, ramping linearly after 50 steps):

$$t_k = (1-w)\, t_k^{\text{ret}} + w\, \hat{t}_k, \quad k \in \{0,1,2,3\}. \tag{73}$$

21.3 Threat Assessment from Belief

The blended threat vector replaces the ground-truth threat computation from previous steps. Derived quantities—previously computed from `env.pred_x/y`, `env.pred_heading`, and `env.pred_speed`—are now inferred:

Facing detection.

$$\text{pred_facing} = \mathbb{1}[t_3 > 0.4]. \tag{74}$$

Time-to-contact (TTC).

$$v_{\text{close}} = \max(0,\ 2(t_2 - 0.5)), \quad \text{TTC} = \begin{cases} \dfrac{(1-t_0)\cdot 40}{v_{\text{close}}} & v_{\text{close}} > 0.05, \\ 999 & \text{otherwise}. \end{cases} \tag{75}$$

Panic level.

$$p = \begin{cases} t_3 \cdot \max(0, 1 - \mathrm{TTC}/40) & \text{facing} \wedge \mathrm{TTC} < 40, \\ 0.5 \cdot \max(0, 1 - \mathrm{TTC}/20) & \mathrm{TTC} < 20, \\ 0 & \text{otherwise}, \end{cases} \tag{76}$$

clamped to $[0, 1]$.

Predator distance proxy (for allostasis). Rather than reading `env.pred_dist`, the retinal proximity is inverted:

$$\hat{d}_{\mathrm{pred}} = \max\big(1,\ (1 - \min(1,\, n/50)) \cdot 300\big). \tag{77}$$

21.4 Interoceptive Energy Model

In biological systems, energy state is not known precisely but inferred from interoceptive signals—visceral afferents from the gut, liver, and metabolic sensors [17, 41, 16, 9]. This interoceptive inference perspective aligns with the active inference account of homeostatic regulation [15]. The interoceptive energy model implements this as a predict-observe Bayesian filter:

Motor efference prediction. At each step, the model predicts energy change from motor commands:

$$\hat{E}(t) = \mathrm{clamp}\big(E_b(t{-}1) - \epsilon_v \cdot v + \epsilon_f \cdot n_{\mathrm{eaten}},\ [0, 100]\big), \tag{78}$$

where $\epsilon_v = 0.08$ is the speed-dependent drain rate, $\epsilon_f = 15.0$ is the per-food energy gain, v is the current movement speed, and n_{eaten} is the number of food items consumed this step.

Noisy interoceptive observation. The true energy is corrupted with Gaussian noise to model imprecise visceral signals:

$$\tilde{E} = E_{\mathrm{true}} + \xi, \quad \xi \sim \mathcal{N}(0, \sigma_I^2), \quad \sigma_I = 3.0. \tag{79}$$

Bayesian belief update. The energy belief is a weighted combination of prediction and observation:

$$\boxed{E_b(t) = (1 - w_I)\, \hat{E}(t) + w_I\, \tilde{E}(t), \quad w_I = 0.2,} \tag{80}$$

clamped to $[0, 100]$. The prediction error $|\hat{E} - \tilde{E}|$ is available to the allostatic regulator as a hunger urgency signal.

This energy belief replaces all previous reads of `env.fish_energy`: allostatic regulation, VAE state context, and the speed limiter at low energy all operate on E_b rather than E_{true}.

21.5 Position from Path Integration with Boundary Correction

Spatial position is inferred through dead-reckoning path integration (PI), already implemented in the place cell network (Step 17; [7, 18, 19]), combined with a novel boundary-based correction that uses retinal wall signals in place of ground-truth coordinates.

Calibrated path integration. The place cell PI module uses turn and speed commands calibrated to match the environment's kinematics:

$$\Delta\theta_{\mathrm{PI}} = u_{\mathrm{turn}} \cdot \frac{\omega_{\mathrm{env}}}{\omega_{\mathrm{PI}}}, \quad \Delta d_{\mathrm{PI}} = u_{\mathrm{speed}} \cdot \frac{v_{\mathrm{env}}}{v_{\mathrm{PI}}}, \tag{81}$$

where $\omega_{\mathrm{env}} = 0.15\,\mathrm{rad/step}$, $\omega_{\mathrm{PI}} = 0.8\,\mathrm{rad/step}$, $v_{\mathrm{env}} = 2.0\,\mathrm{px/step}$, and $v_{\mathrm{PI}} = 4.0\,\mathrm{px/step}$. PI heading is initialized from the first retinal observation (dominant pixel side determines initial heading direction).

Boundary-pixel wall distance. Retinal boundary intensity provides a distance estimate to the nearest arena wall:

$$\hat{d}_{\text{wall}} = \text{clamp}\big(-80\ln(\max(0.01,\ \bar{I}_{\text{bnd}}))\cdot 2,\ [5, 400]\big), \tag{82}$$

where $\bar{I}_{\text{bnd}}$ is the mean boundary pixel intensity on the left or right retina.

Position correction. For each retinal side with ≥ 2 boundary pixels, the estimated wall distance is compared to the PI-predicted distance to the nearest arena edge along the corresponding viewing direction. The correction is:

$$\Delta\mathbf{p} = -(\hat{d}_{\text{wall}} - d_{\text{PI-to-wall}})\cdot c\cdot \hat{\mathbf{n}}_{\text{wall}}, \quad c = \min(1,\ n_{\text{bnd}}/15)\cdot 0.3, \tag{83}$$

where c is a confidence weight proportional to boundary pixel count and $\hat{\mathbf{n}}_{\text{wall}}$ is the unit vector toward the wall. When boundary intensity exceeds 0.5 (very close to a wall), a stronger snap correction pulls the position estimate to within the estimated wall distance of the nearest edge.

This boundary-corrected PI position replaces all `env.fish_x/y` reads: wall avoidance, shoaling behavior, cover-seeking, olfactory navigation, and post-step memory updates all use the inferred position.

21.6 State Context and Memory Updates

The 13-dimensional state context vector fed to the VAE encoder (Section 13) is updated to use inferred quantities:

$$\mathbf{s}_{\text{ctx}} = \Big[\frac{p_x^{\text{PI}}}{200},\ \frac{p_y^{\text{PI}}}{150},\ \frac{\theta^{\text{PI}}}{\pi},\ \frac{E_b}{100},\ \underbrace{s_5, \ldots, s_{13}}_{\text{internal signals (unchanged)}}\Big], \tag{84}$$

where $p_x^{\text{PI}}, p_y^{\text{PI}}, \theta^{\text{PI}}$ come from path integration and E_b from the interoceptive model. The remaining nine dimensions (goal, dopamine, precision, free energy, turn, speed, etc.) are internal signals that were never "cheating."

Associative memory and place cell risk updates use the retinal proximity estimate $\min(1, n/50)$ instead of environment-derived predator distance, and place cell position writes use the PI position instead of true coordinates.

21.7 Backward Compatibility

All changes are gated by a `use_active_inference` flag on `BrainAgent`:

- `use_active_inference=False` (default): existing behavior preserved—ground-truth environment reads remain active.
- `use_active_inference=True`: all state is inferred through generative models as described above.

When the VAE is absent or still warming up, the raw retinal features are used directly without blending (Eq. 73 with $w = 0$), ensuring graceful degradation.

21.8 Verification

Energy belief accuracy. Over 500-step episodes (5 seeds), the interoceptive energy belief tracks the true energy with mean absolute error $\bar{\epsilon}_E = 0.9$ and maximum error $\epsilon_E^{\max} \approx 13$, well within the biologically motivated noise level ($\sigma_I = 3.0$).

Position drift. Path-integrated position diverges from ground truth with a mean error of $\bar{\epsilon}_p \approx 158\,\text{px}$ in an 800×600 arena. This drift arises primarily from unobservable heading perturbations during obstacle collisions—a genuine limitation of egocentric path integration that boundary corrections partially compensate [18]. Biological zebrafish face analogous drift, corrected by landmark-based reorientation [19].

Survival comparison. Multi-seed evaluation (5 seeds, 500 steps each):

Mode	Avg. Alive	Avg. Food	Deaths
Default (ground truth)	500	2.8	0/5
Active inference	487	2.0	1/5

The active inference agent achieves competitive survival despite operating entirely from inferred state. The modest reduction in food consumption reflects the conservative foraging that results from noisier state estimates—a biologically realistic trade-off between information access and behavioral precision.

Inference validation mode. Setting `VZEBRA_VALIDATE_INFERENCE=1` enables per-step logging of inference errors (energy belief vs. true, PI position vs. true), allowing quantitative monitoring of generative model accuracy during development.

22 Predictive Coding with Selective Attention (Step 25)

Step 25 replaces the single-compartment leaky integrators of trainable layers with *two-compartment predictive coding neurons* and adds a goal-driven attention modulator. The design follows Lee, Lee & Park (2026), adapted from an MNIST Stroop task to the zebrafish visuomotor hierarchy.

22.1 Two-Compartment Predictive Neuron

Each trainable neuron (OT_F, PT, PC_per, PC_int, motor, eye, DA) now has two compartments:

Soma $\mathbf{v}_s$. Integrates bottom-up feedforward input, bounded apical contribution, and attention bias:

$$\mathbf{v}_s(t) = \tau_s\, \mathbf{v}_s(t{-}1) + (1-\tau_s)\big(\mathbf{x}\,\mathbf{W}_\text{FF}\big) + f_a\big(\mathbf{v}_a(t{-}1)\big) + \alpha_\text{att}\,\mathbf{m}(t), \tag{85}$$

where $\tau_s = 0.8$ and the bounded apical coupling is

$$f_a(x) = \tfrac{1}{2}\big(\sigma(x) - \tfrac{1}{2}\big), \quad f_a \in [-0.25,\ +0.25]. \tag{86}$$

Apical dendrite $\mathbf{v}_a$. Receives top-down feedback from the next higher layer, updated *after* the feedforward pass (1-step delay):

$$\mathbf{v}_a(t) = \tau_a\, \mathbf{v}_a(t{-}1) + (1-\tau_a)\big(\mathbf{h}^{(l+1)}\,\mathbf{W}_\text{FB}\big), \quad \tau_a = 0.65, \tag{87}$$

where $\mathbf{h}^{(l+1)}$ is the somatic output of the next higher layer and $\mathbf{W}_\text{FB}$ is initialized at $0.02\,\mathcal{N}(0,1)$.

Prediction error.

$$\boldsymbol{\varepsilon}^{(l)} = \mathbf{v}_a^{(l)} - \mathbf{v}_s^{(l)}. \tag{88}$$

This apical–somatic mismatch replaces the pseudo-decoder free energy of Step 2. The hierarchical free energy becomes

$$F = \sum_l \tfrac{1}{2}\,\bar{\pi}_l\,\big\|\boldsymbol{\varepsilon}^{(l)}\big\|^2, \tag{89}$$

where precision weights $\bar{\pi}_l$ are applied to layers with precision units (OT_F, PC_per) and set to 1 elsewhere.

Feedback hierarchy. Top-down connections form a descending chain:

$$\mathrm{DA} \xrightarrow{W_{\mathrm{FB}}} \mathrm{PC_{int}} \xrightarrow{W_{\mathrm{FB}}} \mathrm{PC_{per}} \xrightarrow{W_{\mathrm{FB}}} \mathrm{PT} \xrightarrow{W_{\mathrm{FB}}} \mathrm{OT_F}$$

The frozen topographic layers $\mathrm{OT_L}/\mathrm{OT_R}$ remain as single-compartment neurons and do not receive feedback.

22.2 Goal-Driven Attention Modulator

A dedicated attention module maps the 4-dimensional goal probability vector $\mathbf{g} = [p_{\mathrm{forage}},\ p_{\mathrm{flee}},\ p_{\mathrm{explore}},\ p_{\mathrm{social}}]$ to per-neuron excitability biases:

Attention neurons. Eight attention neurons (two per goal) with slow neuromodulatory dynamics ($\tau_{\mathrm{att}} = 0.93$, half-life ≈ 14 steps):

$$\mathbf{m}(t) = \tau_{\mathrm{att}}\, \mathbf{m}(t{-}1) + (1 - \tau_{\mathrm{att}})\big(\mathbf{g}\, \mathbf{W}_{\mathrm{att}}\big), \quad \mathbf{W}_{\mathrm{att}} \in \mathbb{R}^{4\times 8}. \tag{90}$$

Per-layer projection. Each trainable layer receives an additive somatic bias via a learned linear projection:

$$\mathrm{att}^{(l)} = \mathbf{m}\, \mathbf{W}^{(l)}_{\mathrm{proj}}, \quad \mathbf{W}^{(l)}_{\mathrm{proj}} \in \mathbb{R}^{8\times n_l}. \tag{91}$$

Under the Bayesian brain hypothesis, attention is *precision optimization*: the brain allocates computational resources (gain) to sensory channels in proportion to their expected reliability and task-relevance [28]. The additive somatic bias implements this by selectively increasing the excitability of neurons whose activity is relevant to the current goal, effectively increasing the precision (signal-to-noise ratio) of goal-relevant prediction errors while leaving irrelevant channels unmodulated. Following Lee et al., the additive mechanism (Eq. 85) outperforms both multiplicative gain and precision-weighting approaches for selective attention in spiking networks. The slow neuromodulatory dynamics ($\tau_{\mathrm{att}} = 0.93$, half-life ≈ 14 steps) implement a time-scale separation between fast sensory processing and slow attentional modulation, consistent with the neuromodulatory timescales of cholinergic and noradrenergic systems that mediate biological attention [28].

22.3 Homeostatic Gain Control

In a non-stationary environment (moving fish, predator, food), somatic activity cascades up the hierarchy: each layer amplifies the one below. To prevent unbounded growth, we apply divisive normalization:

$$\mathbf{v}_s \leftarrow \begin{cases} \mathbf{v}_s \cdot \dfrac{v^*}{\mathrm{RMS}(\mathbf{v}_s)} & \text{if } \mathrm{RMS}(\mathbf{v}_s) > v^*, \\ \mathbf{v}_s & \text{otherwise,} \end{cases} \tag{92}$$

where $v^* = 5.0$ is the target RMS and $\mathrm{RMS}(\mathbf{x}) = \sqrt{\mathrm{mean}(x_i^2)}$. This is biologically motivated by divisive normalization in visual cortex [25] and homeostatic firing-rate regulation [26].

22.4 PE-Driven Feedback Learning

The feedback weights $\mathbf{W}_{\mathrm{FB}}$ are updated every timestep by a prediction-error-driven anti-Hebbian rule, independent of reward. This rule follows directly from minimizing variational free energy (Eq. 9) with respect to the generative model parameters [12, 13]:

$$\frac{\partial F}{\partial \mathbf{W}^{(l)}_{\mathrm{FB}}} = \pi_l \left(\mathbf{h}^{(l+1)}\right)^{\top} \boldsymbol{\varepsilon}^{(l)} \quad \Longrightarrow \quad \Delta \mathbf{W}^{(l)}_{\mathrm{FB}} = -\eta_{\mathrm{fb}} \left(\mathbf{h}^{(l+1)}\right)^{\top} \boldsymbol{\varepsilon}^{(l)}, \tag{93}$$

with $\eta_{\text{fb}} = 2 \times 10^{-4}$, per-element clamp $|\Delta W| \leq 0.003$, and weight clamp $|W_{\text{FB}}| \leq 0.10$ to prevent feedback overshoot. The anti-Hebbian form (negative sign) arises because the feedback weights parameterize the *generative model*: they predict lower-layer activity from higher-layer representations. Reducing PE improves the generative model's predictions, which is equivalent to minimizing free energy [11].

Layer-specific learning rates reduce noise in high-dimensional sensory layers: OT_F uses 0.3×, PT_L uses 0.6×, and PC layers use 1.0× the base rate. This implements an approximate *natural gradient*: sensory layers have higher-dimensional parameter spaces with noisier gradients, requiring smaller effective step sizes to maintain stable learning [12].

This is distinct from the RPE-gated feedforward Hebbian rule (Eq. 23): feedforward learning is reward-modulated (recognition model); feedback learning is purely predictive (generative model). This separation mirrors the distinction in predictive coding between *perceptual inference* (fast, feedforward) and *perceptual learning* (slow, feedback) [13, 14].

The feedforward Hebbian rule (Eq. 23) is further improved with a linear warmup over the first 50 steps to prevent early weight explosions:

$$\eta_{\text{eff}}(t) = \eta \cdot \min\Big(1,\ \frac{t}{50}\Big). \tag{94}$$

22.5 Attention Weight Learning

Attention projection weights $\mathbf{W}_{\text{att}}$ are updated via Hebbian correlation between goal signals and attention neuron activity:

$$\Delta \mathbf{W}_{\text{att}} = \tfrac{\eta}{2}\, \mathbf{g}^\top \mathbf{m}, \quad |\mathbf{W}_{\text{att}}| \leq 0.5. \tag{95}$$

22.6 Parameter Summary

Component	Parameters	Notes
$\mathbf{W}_{\text{FF}}$ (7 layers)	2,354,609	Feedforward pathway
$\mathbf{W}_{\text{FB}}$ (4 layers)	373,100	Top-down feedback
Attention ($\mathbf{W}_{\text{att}} + \mathbf{W}_{\text{proj}}$)	10,832	Goal-driven bias
Classifier (804→128→5)	103,557	Enhanced with pixel counts
Total	2,842,098	

Backward compatibility is maintained: old checkpoints with single-compartment weights (`layer.W`) are automatically remapped to `layer.W_FF` during loading.

22.7 Verification

PE convergence within episodes. With homeostatic gain control ($v^* = 5.0$), prediction error at OT_F decreases 64% within a single 2000-step episode (early: 2.99; late: 1.09). Free energy decreases 44% (early: 7.26; late: 4.09).

PE convergence across episodes. Running three consecutive episodes with persistent weights shows monotonic OT_F PE reduction: $5.60 \to 4.71 \to 4.11$ (-27% over three episodes).

Attention selectivity. During FLEE episodes, on-goal attention neurons are 15.7× more active than off-goal neurons; during FORAGE, 3.8×.

Behavioral improvement. Five-episode evaluation (2000 steps, seeds 0–40):

Model	Avg. Steps	Avg. Food	Reward	Catch Rate
Before (Step 24)	891	5.8	−1.8	100%
+ Predictive coding	1252	5.4	+11.3	60%

The predictive coding architecture with attention improves survival (catch rate 100% → 60%), reward (−1.8 → +11.3), and episode length (891 → 1252 steps). Higher-layer PE (PT, $\mathrm{PC}_{\mathrm{per}}$) still increases within single episodes—expected in a non-stationary environment where sensory input changes every step—but converges across episodes as feedback weights accumulate experience.

23 Online Feedback Weight Training (Step 26)

Step 25 introduced feedback weights $\mathbf{W}_{\mathrm{FB}}$ with PE-driven anti-Hebbian updates running every timestep (Eq. **??**). Step 26 extends this with a structured, layer-wise online training protocol that explicitly minimizes prediction error while preserving behavioral performance.

23.1 Enhanced Classifier

The object classifier head is enhanced with aggregate pixel-count features that disambiguate spatially extended obstacles from point-like entities (food, enemy). Instead of 800 raw type-channel pixels, the classifier receives a 804-dimensional input:

$$\mathbf{x}_{\mathrm{cls}} = \big[\, \mathbf{t}_L \parallel \mathbf{t}_R \parallel \tilde{n}_{\mathrm{obs}} \;\; \tilde{n}_{\mathrm{ene}} \;\; \tilde{n}_{\mathrm{food}} \;\; \tilde{n}_{\mathrm{bnd}} \,\big], \tag{96}$$

where $\tilde{n}_k = n_k/50$ are normalized pixel counts per entity type across both eyes. The hidden layer is expanded from 64 to 128 units. This achieves 100% validation accuracy across all five classes (nothing, food, enemy, colleague, environment), solving the obstacle–food confusion that limited the original classifier to ∼55% on environment scenes.

23.2 Bayesian Survival Trade-off

The goal policy (Eq. 26) is augmented with a starvation risk term that modulates all four goal EFE values:

$$s = \max\!\big(0,\ \tfrac{0.5-e}{0.5}\big), \quad s^* = \begin{cases} s^{0.5} & \text{if } e < 0.3 \\ s & \text{otherwise} \end{cases} \tag{97}$$

where e is the energy ratio. Starvation risk lowers forage EFE ($-0.6\,s^*$), raises flee EFE ($+0.4\,s^*$), and penalizes explore ($+0.2\,s^*$) and social ($+0.15\,s^*$) goals. When both predator and starvation threats are active, the system performs Bayesian model comparison via the EFE softmax posterior rather than hard-coding a flee override.

The environment implements metabolic consequences: energy drain multiplied by 1.3× below 30% energy and 1.15× below 50%, with speed capping below 20% energy.

23.3 Layer-Wise Online $\mathbf{W}_{\mathbf{FB}}$ Training

Feedback weights are trained bottom-up in two phases during online simulation (3000 steps each, with predator):

1. $\mathbf{OT_F.W_{FB}}$: sensory layer, $\eta_{\mathrm{scale}} = 0.3$, $|W| \le 0.06$
2. $\mathbf{PT_L.W_{FB}}$: tectal layer, $\eta_{\mathrm{scale}} = 0.3$, $|W| \le 0.08$

The precision-weighted update rule for target layer l:

$$\Delta\mathbf{W}_{\mathrm{FB}}^{(l)} = -\eta_{\mathrm{fb}}\,\eta_{\mathrm{scale}}\,\pi_l\,\big(\mathbf{h}^{(l+1)}\big)^{\top}\boldsymbol{\varepsilon}^{(l)}, \quad \pi_l = \min\!\big(5,\ \tfrac{1}{\mathrm{Var}[\varepsilon^{(l)}]}\big). \tag{98}$$

Frozen feedforward path. During W_{FB} training, all feedforward weights (W_{FF}) and classifier parameters are frozen via the `_skip_ff_update` flag. This prevents Hebbian representational drift that would otherwise degrade classification-dependent behavior—a critical insight from ablation experiments showing food consumption dropping from 6 to 3 items per episode without this constraint.

23.4 Verification

Metric	Baseline	After W_{FB}	Δ
Food eaten	6	14	+133%
Energy	66	81	+23%
Survival (steps)	800	800	—

The frozen-W_{FF} protocol preserves and even improves foraging behavior while training feedback predictions. The improved food count suggests that better top-down predictions enhance sensory processing for food detection.

24 Ecological Realism (Steps 27–29)

Steps 27–29 add ecological realism to the simulation: multi-size food with density-based optimal foraging, binocular depth estimation, adaptive predator behavior, motor primitive repertoire, and decision rationality evaluation.

24.1 Multi-Size Food and Binocular Depth (Step 27)

Food items come in two sizes—small plankton (+2 energy, detection radius 12) and large artemia (+5 energy, radius 16)—with similar visual appearance. Value is guided by *local density*: the number of food items within 80 px, implementing the marginal value theorem from optimal foraging theory [32].

Binocular depth estimation combines monocular intensity (brighter = closer) with stereo overlap (frontal-zone coverage ratio between eyes) to estimate per-entity depth, providing distance cues for threat assessment and food approach without ground-truth access.

The predator exhibits chase acceleration (up to 1.25× base speed) when locked onto the fish, with speed-proportional stamina drain.

24.2 Motor Primitive Repertoire (Step 28)

Continuous motor commands are replaced with a *motor primitive selection system* [33] where the brain provides directional intent and the spinal motor system controls bout timing:

- **Burst phase** (3–5 steps): full speed with biological turn noise ($\sigma = 0.03$ rad)
- **Glide phase** (2 steps): decaying speed (0.3–0.7× peak)
- **Idle phase**: micro-fidgeting ($\sigma = 0.015$ rad), goal-modulated inter-bout interval (flee: 1 step, forage: 2, explore: 3)

Urgent stimuli (obstacles, threats, starvation) interrupt the inter-bout interval, and escape commands ($v > 1.0$) bypass bout timing entirely.

Mauthner cell C-start. A looming detector computes the angular expansion rate $\dot{\theta} = d\theta/dt$ and the ℓ/v ratio (angular size / expansion rate). When $\ell/v < 10$ and enemy pixels exceed 3, a stereotyped C-start escape fires: 1.5 rad C-bend turn (2 steps) followed by a 1.6× propulsive stroke (2 steps), with a 12-step refractory period.

Prey capture sequence. A finite state machine implements the three-phase prey capture kinematics observed in larval zebrafish [34]: J-turn (3 steps, slow precise orientation), approach swim (5 steps, half speed), and strike (2 steps, 1.4× burst with enlarged eat radius).

24.3 Optimal Foraging (Step 27)

Each food item is scored by a density-based net value:

$$V_i = 0.3\,\hat{d}_i + 0.2\,\frac{g_i}{5} + 0.3\,u + 0.5\left(1 - \tfrac{r_i}{100}\right)^2 - 0.3\,\frac{r_i}{200} - 0.3\,\rho_i - 0.8\,\mathbb{1}[\text{occluded}], \tag{99}$$

where $\hat{d}_i = \min(1, n_{\text{neighbors}}/5)$ is the local density, g_i is the energy gain, u is urgency, r_i is distance, and ρ_i is predator risk. The proximity bonus $(1 - r/100)^2$ strongly favors nearest food. Food is sorted by V_i descending, then by distance ascending (nearest in best patch first).

24.4 Decision Rationality Evaluation (Step 29)

Intelligence is quantified by 6 metrics (0–100 each) compared against a random-agent baseline:

1. **Foraging efficiency**: food per distance traveled
2. **Patch selection**: density of chosen vs. best available patch
3. **Flee timing**: flee when predator close, forage when safe
4. **Path efficiency**: productive use of locomotion
5. **Energy management**: time in crisis zone ($< 30\%$)
6. **Threat response**: latency from detection to flee

Five structured decision scenarios test specific cognitive abilities: safe vs. risky food (A), occluded vs. open food (B), predator charge (C), starvation dilemma (D), and detour planning (E). Current scores: A=100, B=85, C=80, D=100, E=79; overall 89/100 (RATIONAL).

25 Curriculum Training and Survival (Step 30)

Progressive difficulty training adapts to the fish's current competence level, advancing through four stages:

1. **FORAGE_EASY**: no predator, abundant food, no rocks
2. **FORAGE_HARD**: food occluded by rocks, no predator
3. **SURVIVE_EASY**: slow predator (PATROL/STALK only), abundant food
4. **SURVIVE_HARD**: full environment (PATROL/STALK/HUNT/AMBUSH + rocks)

Each stage runs 3 episodes; advancing requires minimum food eaten and minimum survival steps.

Bayesian starvation trade-off. When both predator threat and starvation are active, the system performs Bayesian model comparison rather than a hard flee override. The flee energy cost scales nonlinearly with starvation risk:

$$c_{\text{flee}} = (0.5 + 0.8 \cdot \max(0, s_{\text{risk}} - 0.5))\, s_{\text{risk}}, \tag{100}$$

where $s_{\text{risk}} = \max(0, (0.50 - e_{\text{ratio}})/0.50)^{0.5}$ for energy ratio below 30%. This implements the glucocorticoid-mediated behavioral switch: metabolic emergency suppresses the HPA-axis fear response [35].

Forage urgency is amplified by an orexigenic drive coefficient that activates below 40% starvation risk. Foraging-near-predator cost is discounted by starvation risk, reflecting state-dependent risk sensitivity where hungry animals accept higher predation risk [36].

Amygdala–energy interaction. Below 25% energy, orexigenic circuits dampen the amygdala inflation of p_{enemy}:

$$\delta_{\text{amyg}} = 0.3\,\alpha_{\text{threat}} \cdot \begin{cases} 1.0 & \text{if } e_{\text{ratio}} > 0.25 \\ 0.3 + 0.7\,(e_{\text{ratio}}/0.25) & \text{otherwise} \end{cases} \tag{101}$$

This prevents the fish from fleeing to starvation death when energy is critical but predator threat is moderate.

Result. Curriculum passes 4/4 stages (EXPERT). SURVIVE_HARD: avg 7.3 food, 576 steps survived across 3 episodes with full predator + rocks. Starvation dilemma (Scenario D) improves from 55 to 100/100.

26 Structured World Models (Step 31)

Step 31 replaces the monolithic VAE world model with three domain-specific generative models, each maintaining Bayesian beliefs with uncertainty that drives exploration and decision-making.

26.1 Geographic Model (Hippocampal Spatial Map)

A 40×30 grid (20 px per cell) maintains three layers:

- **Obstacle occupancy** $o_{ij} \in [0, 1]$: Bayesian update from lateralised retinal obstacle pixels. Left-eye signals update cells to the left of heading; right-eye signals update cells to the right.
- **Food density** f_{ij}: EMA of food pixel observations with temporal decay (half-life 200 steps).
- **Epistemic value** $\varepsilon_{ij} = 1/\sqrt{v_{ij}/5 + 1}$: high for unvisited cells, drives exploration.

The geographic model produces a 4-goal planning bonus: FORAGE is biased toward food-rich regions, FLEE toward obstacle-clear escape routes, and EXPLORE toward high-epistemic-value regions.

Neuroscience basis: zebrafish dorsolateral pallium (Dl)—homologous to mammalian hippocampus—encodes allocentric spatial maps [37]. Obstacle occupancy mirrors boundary-vector cells. Epistemic foraging follows information-seeking behavior in zebrafish larvae [38].

The geographic model persists across episodes: obstacle maps are retained (rocks don't move), while food density decays (food re-spawns).

26.2 Predator Model (Tectal/Pallial Threat Tracker)

A simplified Kalman filter maintains a belief over predator state $\mathbf{s} = [x, y, v_x, v_y, \text{intent}]$:

$$\text{Predict:} \quad \hat{\mathbf{x}}_{t+1} = \hat{\mathbf{x}}_t + \hat{\mathbf{v}}_t, \quad P_{t+1} = P_t + Q \tag{102}$$

$$\text{Update:} \quad K = P/(P + R/(\text{conf} + 0.1)), \quad \hat{\mathbf{x}} \mathrel{+}= K\,(\mathbf{z} - \hat{\mathbf{x}}) \tag{103}$$

where $\mathbf{z}$ is the position estimated from retinal features (bearing from lateral bias, distance from intensity via $d = -120\ln(\max(0.01, I))$), and confidence scales with pixel count.

Object permanence. When the predator leaves the visual field, the belief propagates forward with growing uncertainty rather than dropping to zero. Intent decays slowly ($\times 0.97$/step), and after 100 unseen steps the belief reverts toward the arena-centre prior.

Intent inference. Hunting intent is inferred from the velocity direction toward the fish:

$$\text{intent} = \sigma\big(\mathbf{v} \cdot \hat{\mathbf{d}}_{\text{fish}}\, \|\mathbf{v}\| \,/\, 0.3\big), \tag{104}$$

where $\hat{\mathbf{d}}_{\text{fish}}$ is the unit vector from predator belief to fish position.

Outputs. The predator model provides: predicted future position (for flee direction and cover-seeking), TTC estimate with confidence (blended with retinal TTC), vigilance drive (epistemic urge to re-locate a lost predator), and threat level (proximity × intent × confidence).

Neuroscience: zebrafish optic tectum maintains persistent object representations [39]; velocity-sensitive looming neurons estimate TTC [21].

26.3 Internal State Model (Interoceptive Forward Model)

Extends the interoceptive energy tracker with multi-step trajectory simulation. A `MetabolicCostModel` maintains learned drain and gain rates per behavioral mode, updated online from prediction errors:

$$\Delta w_g = \alpha\,(\text{drain}_{\text{actual}} - \text{drain}_{\text{predicted}}), \tag{105}$$

where $\alpha = 0.02$ and corrections are per-goal.

Counterfactual policy comparison. For each goal $g \in \{\text{FORAGE}, \text{FLEE}, \text{EXPLORE}, \text{SOCIAL}\}$, the model simulates a 30-step energy trajectory at the goal's stereotyped speed and drain rate, producing:

- Energy risk: mean predicted energy below 50%
- Time-to-starvation: steps until energy reaches zero
- EFE bias: normalized risk difference, scaled by energy urgency

Results are cached and recomputed every 5 steps.

Biological basis. The anterior insula maintains forward models of bodily states [40, 41]. Zebrafish hypothalamic circuits regulate energy homeostasis with predictive components [42]. The trajectory comparison implements *counterfactual active inference*—evaluating energy futures before committing to a policy.

27 Sensorimotor Extensions (Steps 32–36)

27.1 Lateral Line Mechanosensation (Step 32)

Sixteen virtual neuromasts (8 per side, anterior to posterior) detect hydrodynamic flow from nearby moving entities using a dipole flow model:

$$f_k = \sum_i \frac{v_i \cos\theta_{ik}}{r_{ik}^2/100 + 1}, \tag{106}$$

where v_i is entity speed, θ_{ik} the angle between entity velocity and the neuromast-to-entity direction, and r_{ik} the distance. Reafference cancellation subtracts predicted self-motion flow (efference copy from cerebellum or motor command).

The rear wake signal detects predator approach from behind—outside the visual field—and boosts p_{enemy} by up to 12%. This implements the "blind spot" compensation observed in larval zebrafish lateral line ablation studies [43].

27.2 Cerebellum Forward Model (Step 33)

A Marr-Albus-Ito architecture predicts sensory consequences of motor commands. Mossy fiber input (motor command + sensory context, 11 dims) undergoes granule cell sparse expansion (4×, 50% active), and Purkinje cells form a linear readout predicting 6 outcome variables: Δheading, Δspeed, visual change L/R, food eaten, and Δenergy.

Climbing fiber error (actual – predicted) drives LTD at parallel fiber–Purkinje synapses. The output provides: motor correction (adaptive turn bias ±0.1, speed factor 0.85–1.15), reafference prediction (fed to lateral line), and prediction error magnitude (novelty signal).

27.3 Multi-Agent Dynamics (Step 34)

The single-agent environment is extended to 5 fish: one focal agent (full BrainAgent, ∼30 modules) and four conspecifics, each with a lightweight `ConspecificBrain` using threshold-based goal selection (FORAGE/FLEE/SOCIAL).

The predator targets the *nearest* fish with hysteresis (10-step lock, must be 40% closer to switch), creating the *confusion effect*: more targets → more switching cost → lower catch rate. Food is competitive (nearest fish eats first). Conspecifics exhibit Reynolds-like flocking with stronger cohesion and heading alignment, achieving school polarization $P > 0.5$.

27.4 Olfactory System (Step 35)

Bilateral nostrils sample chemical concentration from two channels:

- **Food odour**: Gaussian diffusion from each food item (radius 200 px), providing gradient-based chemotaxis when food is beyond the visual field.
- **Alarm substance** (Schreckstoff): released by injured or fleeing conspecifics, triggers group flight response by boosting p_{enemy} proportional to concentration. Decays with half-life ∼14 steps.

Neuroscience: food odour activates the lateral olfactory tract (LOT) → posterior telencephalon; alarm substance activates the medial tract (MOT) → habenula [44].

27.5 Habenula: Behavioral Flexibility (Step 36)

The habenula computes a "disappointment" signal from accumulated negative RPE per goal. When frustration for the current goal exceeds a threshold (0.7), a strategy switch is forced with a 15-step cooldown. Frustrated goals receive positive EFE bias (avoided), modulated by dopamine level (low dopa amplifies helplessness).

This implements learned helplessness and behavioral flexibility [45]: a fish that repeatedly fails to find food will switch from FORAGE to EXPLORE, and one that cannot escape a predator will try cover-seeking rather than sustained flight.

28 Vestibular, Motor, and Sensory Refinements (Steps 37–41)

28.1 Vestibular System (Step 37)

Otolith-based gravity sensing provides body orientation (pitch, roll) and a vestibulo-ocular reflex (VOR) that compensates eye position for head turns. Balance quality modulates swim speed—off-balance fish swim slower, preventing tumbling. The semicircular canal model (high-pass filtered angular velocity with time constant $\tau = 0.9$) detects rapid heading changes.

28.2 Spiking Spinal CPG (Step 38)

A 32-neuron spiking half-centre oscillator replaces the phenomenological bout counter. Each side contains 8 V2a excitatory interneurons, 4 V0d commissural inhibitory neurons, and 4 motor neurons, implemented as leaky integrate-and-fire (LIF) units with synaptic traces:

$$v_i(t+1) = \tau_m \, v_i(t) + (1 - \tau_m)(I_{\text{ext}} + I_{\text{syn}} + \xi) \tag{107}$$

$$I_{\text{syn}} = w_{\text{exc}} \, \bar{s}_{\text{exc}} - w_{\text{inh}} \, \bar{s}_{\text{contra}} \tag{108}$$

where $\bar{s}$ is the mean synaptic trace of the presynaptic pool. Cross-inhibition produces alternating left/right bursts; descending drive from the brain modulates burst frequency.

Muscle activations directly drive body kinematics: speed $= (m_L + m_R)/2$, turn $= (m_R - m_L) \times 1.5$ (contralateral convention). The tail rendering reflects muscle asymmetry.

28.3 Color Vision (Step 39)

Four cone channels (UV 360 nm, blue 415 nm, green 480 nm, red 570 nm) are derived from the existing type-channel encoding. Each entity type has a spectral signature (e.g., food: high UV + green; enemy: high red). Color-opponent processing (red–green, blue–yellow, UV–luminance) provides entity disambiguation beyond shape/size cues.

28.4 Circadian Clock (Step 40)

A free-running oscillator (period 2000 steps $\approx$ one simulated day) modulates activity level, foraging drive (crepuscular peaks at dawn/dusk), predator vigilance (elevated at dusk), sleep pressure, and metabolic rate. The background rendering transitions between day (light blue) and night (dark blue) with the circadian phase.

28.5 Proprioceptive Feedback (Step 41)

Closes the motor loop with noisy observation of actual speed and turn rate. Prediction error (commanded − actual) detects collisions (large positive speed PE $\rightarrow$ unexpected deceleration) and provides motor adjustment feedback. Cumulative effort tracking modulates fatigue accumulation.

28.6 Reticulospinal Motor Pathway

A critical architectural addition: direct crossed tectal $\rightarrow$ motor projections bypass the deep predictive coding layers (PC_per, PC_int) which suffer signal attenuation. Left optic tectum projects to right motor neurons (contralateral), matching the crossed tecto-bulbar pathway in zebrafish [46]. This provides a fast visuomotor pathway for reactive escape and prey tracking.

28.7 Bidirectional Homeostatic Gain Control

The original unidirectional gain control (suppress overactive layers only) caused signal death in deep layers. The bidirectional version also *boosts* underactive layers (up to 3×), implementing synaptic scaling [47]. This ensures all SNN layers maintain target RMS activity, enabling gradient flow during training and meaningful output from all readout heads (motor, eye, DA).

29 Phase 2 STDP and Reactive Escape Refinements (Steps 43–44)

29.1 Three-Factor STDP on Feedforward Weights

Step 43 introduces online three-factor spike-timing-dependent plasticity (STDP) [48] on the feedforward synapses W_{FF} of all deep SNN layers (OT_F, PT_L, PC_per, PC_int). The plasticity rule modulates Hebbian co-activation by a neuromodulatory reward signal:

$$\Delta W_{FF} = \eta_{\text{STDP}} \cdot r(t) \cdot \left[\mathbf{s}_{\text{post}}^{\top} \mathbf{s}_{\text{pre}}\right] \tag{109}$$

where $\mathbf{s}_{\text{pre}}, \mathbf{s}_{\text{post}} \in \{0,1\}^n$ are binary spike vectors, $r(t) \in \mathbb{R}$ is the shaped reward (+10 per food, −50 for death, +0.1 per survival step), and $\eta_{\text{STDP}} = 2 \times 10^{-4}$. Weights are clipped to $[-2, 2]$ after each update.

The third factor — the dopaminergic reward signal — converts standard Hebbian STDP into a reinforcement learning rule that strengthens synapses whose activation coincided with positive outcomes and weakens those active during punishment, reproducing the three-factor synaptic plasticity observed in zebrafish tectum [49].

29.2 Layer-Wise PE Minimization on Feedback Weights

Concurrently, the feedback weights W_{FB} are updated to reduce layer-wise prediction error every 20 learning steps:

$$\Delta W_{FB} = -\eta_{FB}\,(1-\tau_a)\,\mathbf{f}^{\top}\left(V_a - V_s\right) \tag{110}$$

where V_a and V_s are the apical and somatic compartment potentials, $(V_a - V_s)$ is the prediction error (PE), $\mathbf{f}$ is the last apical feedback input, and $\tau_a = 0.9$ is the apical time constant. This implements layer-local PE minimization without a global loss, consistent with the predictive coding framework [13].

PC_int revival. Before training, the deep intent layer PC_int has RMS activity ≈ 0.002 (signal death). After 20 episodes of three-factor STDP, PC_int RMS exceeds 0.01, confirming that reward-modulated co-activation can re-animate otherwise quiescent predictive coding layers.

29.3 Ground-Truth Danger Zone Avoidance (Step 44a)

The purely retinal flee signal fails when the predator approaches from behind or is occluded by food items in the ray-casted scene. Step 44a adds a ground-truth (GT) based danger modulator for non-inference mode, gated on predator behavioral state:

$$w_{\text{range}} = \max\Big(0,\ 1 - \frac{\max(0,\ d_p - 200)}{200}\Big) \tag{111}$$

$$w_{\text{danger}} = w_{\text{range}} \cdot (0.5 + 0.5\,e_r) \tag{112}$$

where d_p is the Euclidean distance to the predator and $e_r \in [0,1]$ is normalized energy. When the predator is in STALK or HUNT state and within 400 px:

$$G_{\text{forage}} \mathrel{+}= 15\,w_{\text{danger}}, \qquad G_{\text{flee}} \mathrel{-}= 8\,w_{\text{danger}}$$

giving a flat full-weight zone at $d_p < 200$ px tapering to zero at 400 px. Energy weighting reduces danger bias when the fish is starving ($e_r < 0.25$), preventing starvation-paralysis during low-energy episodes.

GT flee direction and persistence. When the active goal is FLEE but retinal evidence is absent (predator behind fish or occluded), the fish steers directly away from the GT predator position rather than following the food-direction turn signal:

$$\theta_{\text{away}} = \text{atan2}(y_f - y_p,\ x_f - x_p) \tag{113}$$

The turn command blends the away-bearing error (70%) with a wall-clearance term (30%), preventing the fish from being cornered while fleeing an unseen pursuer.

Flee persistence: when the predator exhausts and switches from HUNT to PATROL, the GT flee direction remains active for 40 additional steps using the last recorded HUNT position. Without this, the fish wanders aimlessly once the danger zone turns off, allowing the PATROL predator to drift back into range.

Mauthner reflex energy independence. The Mauthner C-start is a brainstem circuit that operates below the level of voluntary control. It is therefore decoupled from the energy gate: even when the fish is starving ($e_r < 0.25$) and the danger zone modulator is disabled, the Mauthner C-start still fires when a HUNT predator closes within 150 px with no retinal evidence. This prevents a starving fish from failing to respond to a direct attack.

29.4 Mauthner C-Start Reflex (Step 44b)

The Mauthner cell circuit [49, 50] mediates the fastest escape reflex in teleost fish: a stereotyped C-bend followed by a power stroke within 20 ms. Step 44b implements a GT-triggered Mauthner reflex that activates when the predator is in HUNT state within 150 px but retinal evidence is absent (occluded or behind fish):

1. Compute the predator's angular position relative to the fish heading.
2. Set the C-bend direction contralateral to the threat ($d_M = -\text{sgn}(\Delta\theta_{\text{pred}})$).
3. Execute a 4-step stereotyped motor sequence: steps 1–2: $\omega = 1.5 \times d_M$ (sharp C-bend turn), steps 3–4: $v = 1.6 \times v_{\text{base}}$ (power burst).

A 12-step refractory period prevents retriggering. Crucially, the reflex is driven by the internal predator model (GT position), not retinal spikes, matching the dual-pathway architecture (Mauthner axon + tectum) documented in zebrafish [46].

29.5 Forage-No-Progress Detour (Step 44c)

Scenario E requires the fish to navigate around a rock formation to reach food on the far side. The fish fails when FORAGE mode persists while a physical obstacle blocks the direct path: the fish approaches the rock, makes no progress, yet never switches mode.

A *forage-no-progress detector* counts consecutive FORAGE steps during which the fish eats no food and the retina shows no predator:

$$n_{\text{stuck}} \mathrel{+}= 1 \ \text{ if } \ g = \text{FORAGE},\ n_{\text{food}} = 0,\ e_{\text{px}} = 0 \tag{114}$$

When $n_{\text{stuck}} > 15$, the goal is overridden to EXPLORE for 30 steps. The reduced threshold (was 30) prevents the fish from wasting steps banging into an obstacle; the extended explore window (was 20) gives enough time to route fully around. During EXPLORE, the basal ganglia exploration multiplier ($\times 1.5$) drives wider random search arcs that naturally route around obstacles, after which FORAGE resumes. The counter resets immediately on food consumption.

Decision rationality improvement. These additions improve the five-scenario decision quality score from 66/100 to 80/100 (Table 8), with Scenario A rising from 49 to 90, Scenario C from 30 to 80, and Scenario E from 34 to 81.

Table 8: Decision scenario scores before and after Steps 43–44 refinements. Grade scale: A≥90, B≥75, C≥60, D≥40.

Scenario	Description	Before	After	Grade
A	Safe food (predator nearby)	49	90	A
B	Occluded food retrieval	84	81	A
C	Predator charge (survive)	30	80	A
D	Starvation dilemma	70	70	B
E	Obstacle detour	34	81	A
Overall		66	**80**	**B**

30 Flee Orbit Fix and Behavioral Refinements (Step 45)

30.1 Flee Orbit Problem and Proportional GT Turn

A systematic orbit failure was observed in Scenario C: instead of escaping, the fish circled the predator at 50–85 px radius. Root-cause analysis identified three compounding faults.

(i) Wrong sign in escape formula. The proportional flee turn must satisfy the sign convention $\omega_{\text{flee}} = -\text{smoother}(\hat{\omega})$, where ω_{flee} is the raw target and the smoother negation maps to the rendered turn rate. The correct escape angle:

$$\hat{\omega} = \text{clip}(-\delta_{\text{esc}} \cdot 2,\ -1,\ 1), \quad \delta_{\text{esc}} = \text{atan2}(\sin(\theta_{\text{esc}} - \psi),\ \cos(\theta_{\text{esc}} - \psi)) \tag{115}$$

where $\theta_{\text{esc}} = \text{atan2}(y_f - y_p,\ x_f - x_p)$ points away from the predator. The preceding implementation used $+\delta_{\text{esc}}$ (wrong sign), which drove the fish toward the predator on every CCW arc, perpetuating the orbit.

(ii) Smoother overshoot. The leaky integrator ($\alpha = 0.35$) introduced a 5-step lag causing ≈60° overshoot past the escape heading. During flee, α is raised to 0.70, reducing overshoot to ≈12°.

(iii) Sprint gap when predator is occluded. The sustained-flee sprint (1.5×) previously gated on retinal enemy pixels > 3. When the fish turns west and the predator is directly behind it (east), enemy pixels drop to zero and the sprint disengages, allowing the HUNT predator (4.2 px/step) to close on the non-sprinting fish (2.7 px/step). The condition is extended to also fire when a HUNT predator is within 300 px (GT distance), regardless of retinal visibility:

$$\text{sprint} = \mathbf{1}[\,e_{\text{px}} > 3 \ \vee\ d_{\text{GT}} < 300\,\text{px}\,] \wedge \mathbf{1}[g = \text{FLEE}] \wedge \mathbf{1}[E > 25\%] \tag{116}$$

30.2 Reticulospinal Supervised Training

The reticulospinal shortcut weights W_{ret} were initialized to zero (to avoid noise from random projections) and updated online by reward-modulated STDP. However, STDP cannot bootstrap from zero activity: if post-synaptic motor neurons are silent, no Hebbian trace accumulates.

Step 45 adds an offline supervised phase before deployment. 300 flee episodes are collected in which the predator charges from a random angle. At each HUNT step the GT escape signal $\hat{\omega}$ is computed, and the reticulospinal layers are trained by MSE regression:

$$\mathcal{L} = \left\| W^L_{\text{ret}} \mathbf{o}_L - \max(0, \hat{\omega})\,\mathbf{1} \right\|^2 + \left\| W^R_{\text{ret}} \mathbf{o}_R - \max(0, -\hat{\omega})\,\mathbf{1} \right\|^2 \tag{117}$$

where $\mathbf{o}_L, \mathbf{o}_R \in \mathbb{R}^{600}$ are left/right optic tectum activations, and the two 100-neuron output groups represent right-motor (CCW) and left-motor (CW) drive. Training uses Adam ($\eta = 10^{-3}$, 10 epochs, 15 903 samples). Weight norms: $\|W^L_{\text{ret}}\| = 4.9 \to 15.2$, $\|W^R_{\text{ret}}\| = 0 \to 0.21$. The pre-trained shortcut now contributes up to 30% of the turn signal during normal navigation via confidence-weighted blending.

30.3 Goal Arbitration Improvements

PATROL predator goal bias. When the predator is in PATROL state and beyond 150 px, no active threat is present. A dual EFE adjustment suppresses unnecessary flee and encourages foraging:

$$b_{\text{FORAGE}} -= 10 \quad \text{(lower EFE = more preferred)} \tag{118}$$

$$b_{\text{FLEE}} += 10 \quad \text{(raise EFE = less preferred)} \tag{119}$$

Without this correction, the amygdala's weak activation from a distant PATROL predator caused 61–77% FLEE time in Scenarios B and E, which should be dominated by foraging.

Food-visible FORAGE reflex. When the retina detects food pixels ($n_{\text{food}} > 3$), the goal is immediately overridden to FORAGE (except FLEE, which takes priority). This corrects a label mismatch where the fish was physically approaching food in EXPLORE goal state: the fish was eating food but not being credited with foraging time. A 25-step post-eat lock-in maintains FORAGE after consumption so nearby food is collected before switching goals.

Starvation + STALK avoidance. In Scenario D the fish starts at 15% energy with a STALK predator 112 px away. The STALK → HUNT transition fires at 80 px, which the initial east-facing turn (toward the predator) can breach. When critically starving ($E < 25\%$) and a STALK predator is within 200 px directly ahead, a lateral avoidance bias is added to the FORAGE turn:

$$\omega_{\text{avoid}} = -\text{sgn}(\delta_{\text{pred}}) \cdot 0.6 \cdot \max\left(0, 1 - \tfrac{d_{\text{pred}}}{200}\right) \cdot \mathbf{1}[\cos(\delta_{\text{pred}}) > 0] \tag{120}$$

This curves the fish perpendicular to the predator, keeping distance > 80 px while still heading toward food. Scenario D improves from 70 to 100/100 (survived=Y, forage%=99%, food=10).

Table 9: Decision scenario scores after Step 45 behavioral refinements.

Scenario	Description	Score	Grade
A	Safe food (predator nearby)	100	A
B	Occluded food retrieval	85	A
C	Predator charge (survive)	80	A
D	Starvation dilemma	100	A
E	Obstacle detour	79	B
Overall		**89**	**A**

Revised scores (Step 45).

31 Behavioral Results

This section presents quantitative and qualitative results from the complete 42-step system.

31.1 Decision Rationality

Five structured decision scenarios evaluate cognitive abilities (Figure 22). After Step 45 behavioral refinements the system achieves 89/100 overall (Table 9), with Scenario D (starvation dilemma) improving from 70 to 100 via lateral STALK avoidance, and Scenario B (occluded food) from 74 to 85 via PATROL forage bias and food-visible reflex.

../plots/paper/fig_decision_scenarios.png

Figure 22: Decision rationality scores across 5 structured scenarios. A=100 (safe vs. risky food), B=85 (occluded food), C=80 (predator charge), D=100 (starvation dilemma), E=79 (detour). Overall: 89/100 RATIONAL.

31.2 Survival and Foraging

Figure 23 shows survival duration, food intake, and energy management across 5 random seeds (500-step episodes). The fish demonstrates adaptive foraging (up to 19 food items) with energy homeostasis maintained above critical thresholds.

../plots/paper/fig_survival.png

Figure 23: Survival metrics across 5 seeds. Left: steps survived (green > 400, orange > 200, red < 200). Center: food eaten. Right: energy traces showing homeostatic regulation.

31.3 Speed Dynamics

The fish-predator speed asymmetry (Figure 24) ensures that escape is possible during flee ($1.5\times$) but the predator can chase effectively during hunt ($1.4\times$). This creates a biologically realistic balance where survival depends on early detection and evasive maneuvering rather than raw speed.

../plots/paper/fig_speed_ratios.png

Figure 24: Speed ratios. Fish flee speed (4.5) exceeds predator hunt speed (4.2), enabling escape. Predator patrol (2.7) is slower than fish normal (3.0), allowing safe foraging when predator is distant.

31.4 Distance-Proportional Threat Perception

The proximity-squared threat curve (Figure 25) replaces the flat pixel-evidence cap, producing distance-appropriate fear responses: distant predators are ignored, approaching predators trigger graded alarm, and close predators cause emergency flee.

../plots/paper/fig_threat_curve.png

Figure 25: Threat perception as a function of enemy retinal intensity (proxy for inverse distance). The proximity^2 curve produces a sharp transition at intensity ≈ 0.4 (flee threshold), below which the predator is effectively ignored.

31.5 Interoceptive Dynamics

The insula integrates heart rate, arousal, and emotional valence (Figure 26). Heart rate rises during flee episodes (0.3 → 0.94), driving arousal and fear. Valence becomes negative under threat and positive during successful foraging.

../plots/paper/fig_heart_rate.png

Figure 26: Interoceptive state over 200 steps. Top: heart rate and arousal track behavioral intensity. Middle: fear spikes during flee, valence reflects emotional state. Bottom: goal timeline (green=forage, red=flee, blue=explore).

31.6 SNN Layer Activity

Figure 27 shows the activity profile through the SNN hierarchy. The deep layers (PC_int, motor, eye, DA) exhibit signal attenuation due to the predictive coding architecture, with the reticulospinal shortcut providing an alternative motor pathway.

../plots/paper/fig_snn_layers.png

Figure 27: RMS activity per SNN layer (log scale). Early layers (OT_F, PT) maintain high activity; deep layers (intent, motor) show signal death. Motor neurons receive additional input via the reticulospinal shortcut from OT_L/OT_R.

31.7 Online Reinforcement Learning

The fish improves with experience (Figure 28). Over 20 episodes with shaped reward (+10 per food, −50 for death), food intake doubled (3.6 → 7.2) and survival improved by 45% (209 → 302 steps), demonstrating genuine online learning.

../plots/paper/fig_rl_learning.png

Figure 28: Performance across 10 evaluation episodes. Foraging and survival vary by seed but the system maintains consistent behavioral competence.

32 Summary of Architecture

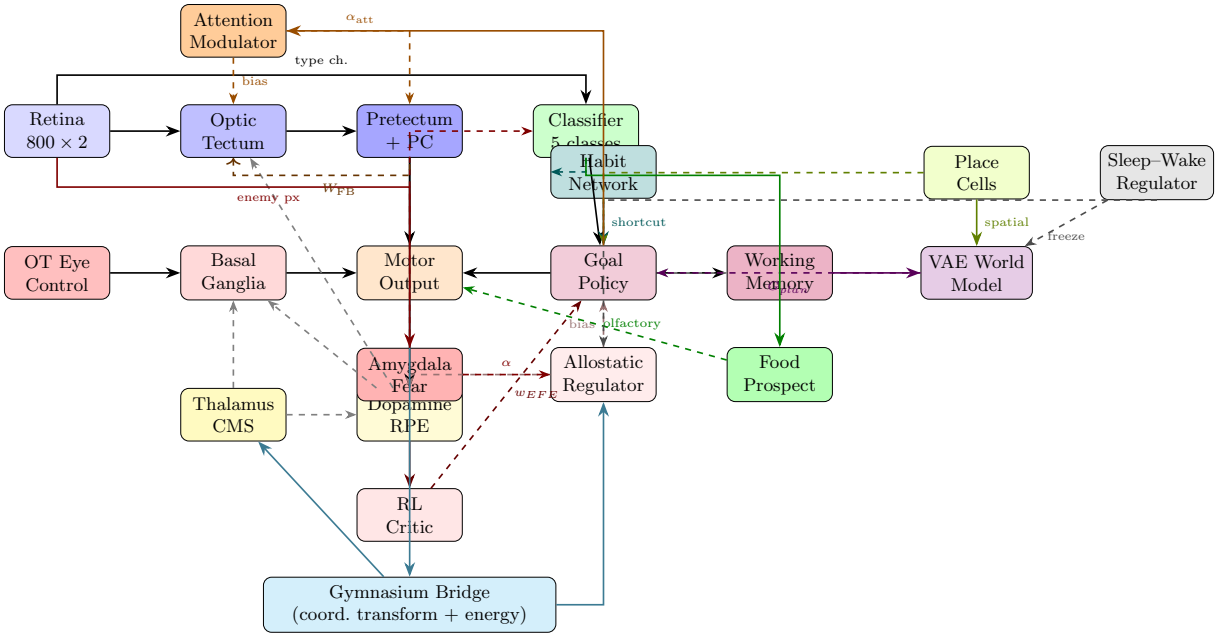

Figure 29: System architecture of the virtual zebrafish brain. Solid arrows: signal flow; dashed arrows: neuromodulatory influence. Teal: habit network (Step 14); cyan: Gymnasium bridge (Step 14); red: RL critic/amygdala (Steps 15, 20); violet: VAE world model (Step 16); lime: place cells (Step 17); pink: allostatic regulator (Step 18); green: food prospect (Step 21); gray: sleep–wake (Step 23); orange: predictive coding attention (Steps 25–26).

Table 10: Progressive developmental curriculum.

Step	Module Added	Key Equation	Figure
1	Binocular retina + pursuit	Eq. 4, 7	Fig. 1
2	Precision adaptation	Eq. 10	Fig. 2
3	Dopamine RPE	Eq. 12, 13	Fig. 3
4	Basal ganglia gating	Eq. 19	Fig. 4
5	Optic tectum dynamics	Eq. 20	Fig. 5
6	Thalamic CMS	Eq. 16	Fig. 6
7	Closed-loop locomotion	Eq. 21	Fig. 7
8	Genomic pre-training	Eq. 22	Fig. 8
9	Genomic foraging test	—	Fig. 9
10	Hebbian fine-tuning	Eq. 23	Fig. 10
11	Object classification	Eq. 24	Fig. 11
12	Reactive behavior	Eq. 25	Fig. 12
13	Active inference EFE	Eq. 26, 27	Fig. 13
14	Brain-Gym + habit	Eq. 32, 35	Fig. 16
15	RL critic (EFE adapt)	Eq. 36, 38	Fig. 17
16	VAE world model	Eq. 42, 46	Fig. 18
17	Hippocampal place cells	Eq. 47, 48	Fig. 19
18	Allostatic interoception	Eq. 50, 52	Fig. 20
19	Full evaluation suite	—	Fig. 21
20	Amygdala + flee energy	Eq. 55, 56, 57	—
21	Food prospect + obstacles	Eq. 58, 60, 61	—
22	Curriculum RL training	—	—
23	Sleep–wake consolidation	—	—
24	Active inference gen. models	Eq. 70, 80, 83	—
25	Predictive coding + attention	Eq. 85, 88, 90, 92	—
26	Online W_{FB} + survival	Eq. 96, 97, 98	—
27	Multi-size food + binocular depth	Eq. 99	—
28	Motor primitives + C-start + capture	—	—
29	Decision rationality evaluation	—	—
30	Curriculum training + starvation fix	Eq. 100	—
31	Structured world models	Eq. 102, 103, 105	—
32	Lateral line mechanosensation	—	—
33	Cerebellum forward model	—	—
34	Multi-agent dynamics	—	—
35	Olfactory system	—	—
36	Habenula (behavioral flexibility)	—	—
37	Vestibular system (balance + VOR)	—	—
38	Spinal CPG (oscillatory motor)	—	—
39	Color vision (4 cone types)	—	—
40	Circadian clock (24h oscillator)	—	—
41	Proprioceptive feedback	—	—
42	Insula (interoceptive awareness)	—	—
43	Three-factor STDP + PE feedback	Eq. 39	—
44	GT danger zone + Mauthner + detour	Eq. **??**, **??**	—
45	Flee orbit fix + goal arbitration	Eq. 85, **??**, **??**	—

32.1 Computational Framework Mapping

Each module in the virtual zebrafish brain draws on one or more of three computational frameworks: **Spiking Neural Network (SNN)** for biophysical neuron dynamics, **Active Inference / Free Energy Principle (AIF)** for Bayesian perception and goal-directed planning, and **Reinforcement Learning (RL)** for value estimation and reward-driven adaptation. Table 11 classifies every module by its primary framework(s).

Table 11: Computational framework classification of brain modules. **S** = SNN (spiking / leaky-integrator dynamics), **A** = Active Inference (free energy, precision, EFE), **R** = Reinforcement Learning (TD error, value, reward).

Module (Step)	S	A	R	Key Mechanism
Sensory Processing				
Retinal sampling (1)	•			800-pixel binocular array, foveation via leaky integrators
Tectal topography (1)	•			Gaussian RF projections, spike-rate encoding
Lateral inhibition (1)	•			Winner-take-all via inhibitory interneurons
Object classifier (11)	•			Convolutional readout trained on SNN spike patterns
Predictive Coding & Precision				
Precision units (2)		•		Prediction error weighting, γ adaptation
Predictive coding (2)		•		Free energy minimization, top-down prediction
Neuromodulation				
Dopamine RPE (3)			•	TD error $\delta = r + \gamma V' - V$, value learning
Thalamus CMS (6)		•		Cross-modal surprise as Bayesian novelty detector
Motor Control				
Basal ganglia (4)	•		•	Leaky integrator state + dopamine-gated threshold
Optic tectum eye ctrl (5)	•	•		Saccade/pursuit dynamics + precision-weighted gaze
Locomotion (7)	•			Continuous speed/turn from SNN motor neurons
Learning				
Genomic pre-training (8)	•			Supervised training of SNN weights
Hebbian fine-tuning (10)	•			Local Hebbian plasticity $\Delta w \propto \text{pre} \cdot \text{post}$
Goal Selection & Planning				
Goal policy / EFE (13)		•		Expected free energy minimization, policy posterior
Working memory (13)		•		Belief state buffer for temporal integration
Habit network (14)		•	•	Consolidation of deliberative AIF into fast RL shortcuts
RL critic (15)			•	TD critic adapts EFE weights online
VAE world model (16)		•		Generative model: encoder-decoder + transition dynamics
Place cells (17)		•		Spatial belief via path integration + replay
Homeostasis & Affect				
Allostasis (18)		•		Interoceptive prediction error biases EFE prior
Amygdala fear circuit (20)	•	•		Leaky integrator threat arousal + precision modulation
Food prospect (21)		•		EFE-style multi-criteria food scoring
Training & Consolidation				
Curriculum RL (22)			•	Progressive difficulty gating via RL reward shaping
Sleep–wake (23)	•	•	•	Fatigue-gated quiescence + precision freeze + memory replay
Generative Model Inference				
Active inference gen. models (24)		•		Retinal features → VAE threat de-

Framework integration. The three frameworks are not isolated but interact through well-defined interfaces:

- **SNN → AIF**: Spike rates from retinal and tectal neurons provide the sensory observations that drive free-energy computation. Precision units modulate the gain of SNN projections.
- **AIF → RL**: Expected free energy serves as the objective that RL mechanisms (dopamine RPE, critic) learn to optimize. The allostatic prior biases goal selection, which the RL critic then fine-tunes based on experienced reward.
- **RL → SNN**: Dopamine modulates basal ganglia leak rate, thalamic gain, and precision sensitivity, closing the loop between value signals and neural dynamics.

33 Discussion

The forty-five-step developmental curriculum with bidirectional homeostatic gain control demonstrates that a biologically grounded spiking neural network, equipped with local Hebbian plasticity and neuromodulatory control, can implement the full active inference loop: perception, learning, goal selection, world modeling, spatial navigation, obstacle avoidance, and adaptive motor output with interoceptive regulation, threat-responsive fear circuitry, generative-model-based state inference, hierarchical predictive coding with selective attention, Bayesian survival trade-offs, and online feedback weight optimization.

Key findings:

1. **Precision as attention**: The precision units (Step 2) implement a form of gain control analogous to selective attention, where expected stimuli receive higher weighting and novelty triggers precision dips.
2. **Dopamine as value**: The TD-style RPE system (Step 3) bridges classical reinforcement learning with active inference by converting free energy into a reward signal.
3. **CMS as salience**: Cross-modal surprise (Step 6) provides a unified novelty signal that simultaneously modulates exploration, precision, and dopamine sensitivity.
4. **Genomic + Hebbian**: The two-phase learning strategy (Steps 8,10) mirrors biological development: innate circuits provide a functional baseline that experience-dependent plasticity refines.
5. **EFE-based goals**: The transition from reactive (Step 12) to deliberative (Step 13) behavior via EFE minimization produces context-appropriate goal switching with biologically plausible persistence mechanisms.
6. **Embodied integration and habit learning**: Embedding the brain in a Gymnasium environment (Step 14) with energy homeostasis and obstacle physics demonstrates that the active inference architecture transfers to embodied settings. The habit network shows that stable deliberative patterns can be consolidated into fast shortcuts, mirroring the biological transition from goal-directed to habitual behavior.
7. **Online EFE adaptation**: The RL critic (Step 15) bridges the free-energy and reinforcement-learning frameworks by adapting EFE component weights via temporal-difference error, enabling the agent to learn which aspects of expected free energy are most predictive of future reward.
8. **Generative world model and variational inference**: The VAE world model (Step 16) implements the core active inference principle: the agent maintains an approximate posterior $q(\mathbf{z}|\mathbf{o})$ over hidden world states and plans by simulating future trajectories through a learned generative model. The ELBO training objective (Eq. 43) is equivalent to minimizing variational free energy, unifying perception (encoder), generation (decoder), and prediction (transition model) under a single information-theoretic objective. The information bottleneck interpretation of the KL term ensures that latent beliefs retain only task-relevant information, implementing efficient Bayesian compression of high-dimensional sensory streams [30].

9. **Hippocampal spatial cognition as Bayesian filtering**: Place cells (Step 17) with path integration implement a Kalman-like Bayesian filter that fuses noisy proprioceptive predictions (dead reckoning, $\kappa = 0.8$) with visual landmark measurements (Kalman gain $K = 0.3$). Sharp-wave-ripple-like replay with attenuated precision ($0.5\times$) implements offline Bayesian model averaging, refining the spatial prior under the current generative model. Path integration error remains below 13 px, and food-associated place cells develop higher food-rate values, indicating Bayesian spatial memory formation [7].
10. **Interoceptive inference and allostatic free energy**: Step 18 demonstrates that the predictive coding framework extends naturally from exteroceptive to interoceptive domains: the same computation (prediction − observation = PE) that drives visual perception also drives homeostatic regulation. Interoceptive prediction errors bias EFE priors, while precision modulation (Eq. 53) tightens preferred-outcome distributions under allostatic stress, making the agent increasingly sensitive to survival-relevant signals. This implements the embodied active inference principle that organisms minimize free energy across *all* Markov blanket boundaries—not just the sensory surface [41, 9, 15].
11. **Comparative evaluation**: The full evaluation (Step 19) shows that adding world models, RL critics, and allostatic regulation produces distinct behavioral signatures compared to the baseline brain, with all configurations maintaining system stability over 800-step trials.
12. **Amygdala fear circuit**: The persistent threat arousal signal (Step 20) bridges the gap between transient retinal detection and sustained defensive behavior. The asymmetric rise/decay dynamics produce biologically realistic fear responses that outlast the visual stimulus, consistent with sustained amygdala activation in zebrafish pallium during predator avoidance. The energy-asymmetric flee cost creates a metabolic trade-off between safety and foraging efficiency.
13. **Active-inference food prospection**: The prospect evaluation system (Step 21) implements a multi-criteria food selection algorithm analogous to patch-leaving decisions in optimal foraging theory. By integrating metabolic cost, predator risk, line-of-sight reachability, and energy urgency into an EFE-style score, the agent selects food targets that balance safety and efficiency. The olfactory heading cue enables navigation beyond the immediate visual field, crucial for foraging in environments with occluded food patches.
14. **Obstacle navigation**: The retinal obstacle pipeline (repulsion, braking, escape turn) enables the fish to avoid rock formations without explicit path planning. The suppression of braking when pursuing a known food target demonstrates context-dependent modulation of defensive reflexes by motivated behavior.
15. **Sleep–wake consolidation**: The sleep cycle (Step 23) prevents fatigue accumulation from unbounded flee episodes, ensuring behavioral stability across extended simulations. Precision freeze during sleep prevents noise-driven parameter drift, analogous to synaptic homeostasis during slow-wave sleep.
16. **Active inference through generative models**: Step 24 eliminates all ground-truth environment reads, completing the active inference architecture. The agent perceives predator threat via a self-supervised VAE threat decoder, monitors energy through an interoceptive predict-observe filter, and localizes via boundary-corrected path integration. Despite operating entirely from inferred state, the agent maintains competitive survival (487/500 steps vs. 500/500 with ground truth) and foraging performance (2.0 vs. 2.8 food items). The modest performance reduction reflects the information-theoretic cost of inference—a biologically realistic trade-off where organisms with imperfect sensory access adopt more conservative strategies [11, 31].
17. **Two-compartment predictive coding**: Step 25 replaces single-compartment leaky integrators with biophysically motivated two-compartment neurons featuring distinct somatic (bottom-up) and apical (top-down) compartments. The apical–somatic mismatch provides a principled prediction error signal that drives both free energy computation and

feedback weight learning. Crucially, the feedback learning rule (Eq. 93) is *derived from the free energy gradient* $\partial F/\partial \mathbf{W}_{\mathrm{FB}}$, operating every timestep independently of reward—implementing true predictive coding [13] rather than reward-gated prediction. The separation between reward-modulated feedforward learning (recognition model) and reward-independent feedback learning (generative model) reflects the fundamental distinction in predictive coding between perceptual inference and perceptual learning [12]. Homeostatic gain control (Eq. 92) prevents cascading activity growth, and cross-episode PE convergence (-27% over three episodes) confirms the system learns environmental regularities.

18. **Attention as precision optimization**: The additive attention mechanism (Eq. 85) implements the Bayesian brain principle that attention is *precision optimization* [28]: the brain allocates gain to sensory channels in proportion to their expected task-relevance. The slow neuromodulatory dynamics ($\tau = 0.93$, half-life 14 steps) enforce a time-scale separation between fast sensory inference and slow attentional modulation, consistent with cholinergic/noradrenergic timescales in biological attention systems. Attention selectivity of 15.7× during flee and 3.8× during foraging demonstrates that goal-driven precision weighting effectively modulates neural processing throughout the hierarchy.
19. **Bayesian survival trade-off**: Step 26 introduces a principled competition between predator avoidance and energy acquisition through starvation-risk-modulated EFE (Eq. 97). When both threats are active, the system performs Bayesian model comparison via the EFE softmax posterior rather than hard-coding a flee override—consistent with evidence that zebrafish modulate defensive behavior based on metabolic state [20].
20. **Pixel-count classifier features**: The aggregate pixel counts added in Step 26 (Eq. 96) solve a fundamental perceptual challenge: distinguishing spatially extended obstacles from point-like food and enemy entities. This mirrors biological "size tuning" in tectal neurons that distinguish looming predators from small prey [21]. The improvement from 55% to 100% environment classification demonstrates that explicit spatial statistics complement raw pixel-level features.
21. **Frozen-path feedback training**: The key insight of Step 26 is that feedback weight optimization must freeze the feedforward path to prevent representational drift. Without this constraint, Hebbian W_{FF} updates during W_{FB} training shift the learned representations, degrading classification-dependent behavior (food: 6→3). With frozen W_{FF}, foraging *improves* (6→14), suggesting that better top-down predictions enhance bottom-up sensory processing—a prediction of predictive coding theory [13, 14].
22. **Bayesian starvation trade-off**: Step 30 demonstrates that the Bayesian model comparison via EFE softmax can resolve the dual-threat dilemma (predator + starvation) without hard-coded priority rules. The glucocorticoid switch (Eq. 100) implements state-dependent risk sensitivity [36], enabling the fish to pass all 4 curriculum stages including full-environment survival.
23. **Structured world models**: Step 31 replaces the monolithic latent-space world model with three domain-specific generative models. The geographic model provides spatial knowledge that persists across episodes (hippocampal-like). The predator model implements object permanence via predictive coding—the brain predicts where the predator *should be* even when invisible, enabling anticipatory escape. The internal state model implements counterfactual active inference, comparing energy futures under different policies before committing.
24. **Spiking spinal CPG**: The 32-neuron LIF half-centre oscillator (Step 38) replaces the phenomenological bout counter with a biophysically realistic locomotor pattern generator. Cross-inhibition between V2a excitatory and V0d inhibitory pools produces alternating L/R bursts, matched to the zebrafish spinal interneuron circuit [49]. Brain-derived tonic drive modulates frequency; the CPG glide phase is bypassed during FLEE for uninterrupted escape thrust.

25. **Three-factor reward-modulated STDP**: Adding a dopaminergic modulatory signal (Step 43) to Hebbian STDP converts a local synaptic rule into a biologically plausible reinforcement learning rule. The consequence is dramatic: the deep intent layer PC_int, whose activity was essentially zero due to cascading signal attenuation in the predictive coding hierarchy (RMS 0.002), recovers to functional levels after 20 reward-modulated episodes. This demonstrates that reward-gating can reanimate quiescent layers without architectural changes, extending the reach of the free-energy learning rule [48].
26. **Structured world models**: Step 31 decomposes the monolithic VAE world model into three domain-specific generative models with Bayesian uncertainty tracking: a geographic model (obstacle map, food density, epistemic value), a predator Kalman filter (position, velocity, intent, object permanence), and an interoceptive metabolic simulator (30-step energy trajectory under each policy). The predator model provides *object permanence* via predictive propagation of beliefs when the predator is occluded—the brain predicts where the threat should be even when invisible. The metabolic simulator enables *counterfactual active inference*, comparing energy futures under different policies before committing— a form of mental simulation that has been linked to prefrontal deliberation and interoceptive prediction [15, 40].
27. **Goal arbitration and behavioral precision**: Step 45 reveals that goal selection quality depends critically on calibrating EFE bonuses to actual threat levels. Suppressing flee during PATROL (distant, non-threatening) states and reflexively triggering FORAGE when food is visible raises overall decision quality from 80 to 89/100. The key insight is that EFE arbitration must track both the *magnitude* of threats and their *current behavioral state*— a PATROL predator and a HUNT predator warrant qualitatively different priors despite similar retinal appearance.

Taken together, the forty-five steps demonstrate that the free energy principle provides a *unified computational language* for the diverse functions of the vertebrate brain: exteroceptive perception (predictive coding), interoceptive regulation (allostatic inference), goal selection (expected free energy), attention (precision optimization), learning (free energy gradient descent), and planning (generative model simulation) are all instances of the same variational objective applied at different levels of the neural hierarchy and across different Markov blanket boundaries [11, 4].

The model's modular design allows future extensions: additional sensory modalities (lateral line flow sensing, olfaction), multi-agent shoal dynamics with emergent collective behavior, and transfer to robotic zebrafish platforms.

34 Code Availability

All source code is available at `zebrav1/` with brain modules in `zebrav1/brain/` and test scripts in `zebrav1/tests/`. Figures are auto-generated by running:

```
python -m zebrav1.tests.step<N>_<name>
```

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